Lyrebird: a physically interpretable syllable inventory for wild bird recordings, and the measured limits of what the archives support

[TODO: author name, affiliation, ORCID]

1 August 2026

Abstract

Lyrebird measures syllables in wild bird recordings harvested from xeno-canto and iNaturalist, fits the Laje–Mindlin normal-form syrinx oscillator to each measured syllable type by coordinate descent, learns a per-species type inventory and a first-order grammar over it, and ships the result as a browser application in which live or synthesised sound illuminates the acoustically nearest regions of a three-dimensional map. The corpus is 340 079 measured syllables from 10 184 recordings, and the fit covers 2 421 species and 13 859 syllable types. Recordings are decoded, described as numbers and deleted, so no audio is redistributed and no model weights reach the browser. Every claim the interface makes carries one of five evidence labels. We report what the corpus supports and where it stops. A first-order transition model over the learned token inventory buys 0.98 bits per token on held-out bouts while costing 0.75 bits on a within-species shuffle of the same tokens (34 696 bouts, 327 164 tokens, 2 421 species) — an unbiased estimator paying for parameters it cannot use is the clean form of the claim that the tokens are units. A second order is not supported: going from order 1 to order 2 the median species loses 0.091 bits per token, and only 217 of 2 421 species have enough bouts to ask. Adding repeat classes to the song detector takes the corpus from 48 to 98 species with a measured song while the shuffle control stays at 0.170 % of candidate bouts. Three results run against the design. The scalar the pipeline optimises — a cosine in a 28-dimensional descriptor space — carries essentially no information about audible spectral distortion ($r \approx 0.1$ against two metrics that agree with each other at 0.925, $n = 200$ types). The learned embedding organises by recordist before it organises by bird (nearest-neighbour lift +0.488 against +0.196), and four independent attempts to remove that confound failed, including two archive-scale external models that disagree with each other by more than either differs from ours. And with every type granted its own cluster, the share of syllables whose nearest centroid belongs to their own type is 0.602, which is the honest ceiling on "sing and the right region lights". Appendix A lists all 58 pre-registered gates and their outcomes; 20 passed, 36 ended in a written refusal, and 2 are unrun. The refusals are as much of the contribution as the system is.

Keywords: bioacoustics; birdsong; syrinx model; gesture inversion; contrastive embeddings; sequence statistics; negative results; pre-registration.

The software repository is named `Birdy`; the system is called Lyrebird here and in all subsequent writing. Internal provenance for every number in this paper is given as a comment in the LaTeX source, naming the research note and section that measured it.

1 Premise

Lyrebird is an instrument in three parts.

It *sings* from physics. Every note is integrated at audio rate from the low-dimensional normal-form oscillator of the Laje–Mindlin lineage [24, 40, 42, 46], driven by the two scalar gestures a real bird controls: air-sac pressure and syringeal tension. Nothing is sampled, no waveform is replayed, and no neural generator is involved. The parameters that produce a given note are the parameters a physiologist would name.

It *remembers* without keeping audio. The harvester fetches a recording, decodes it, segments it into syllables, describes each syllable as a 28-dimensional acoustic descriptor and a 32×23 log-mel patch, and deletes the file. The production store holds 340 079 such measurements over 10 184 recordings and 3 811 species, of which 2 421 species carry a fitted inventory of 13 859 syllable types. The measurements are not invertible to recognisable audio, so the store is a description of an archive rather than a copy of one.

It *shows* the result as a map. One point is one measured syllable. Points are placed by projecting the space in which syllable identity is actually decided — a learned 32-dimensional embedding — down to three dimensions, and clusters in that space are labelled with the syllable type, the documented behavioural function where one exists, and the citation for it. When a microphone hears a sound, or when the synthesiser sings one, the regions of the map nearest to it in the acoustic space light up.

What the system claims about meaning is narrow and is printed on its face. It maps a call type to a documented behavioural function, and it attaches to every phrase it speaks one of five labels: *experimental* (function established by playback experiment), *documented* (function described in the literature), *inferred* (meas-

ured in this corpus, function not established), *invented* (a designed mapping, named as designed), and *unlabelled* (real acoustics, no documented function). Playback is the only proof of function the field accepts [65]; Lyrebird has run none of its own, so everything past the literature is labelled as inference or as invention.

This paper reports the instrument and, at greater length, the measurements that constrain it. The project's working discipline is that a change to fitting, to an objective or to a space is judged by a metric written down before the run, on a fixed corpus, and that a result which contradicts the plan is recorded in place rather than deleted. Fifty-eight such gates have been run. Thirty-six of them failed. Sections 7 to 9 are organised around what survived and what the failures rule out.

2 Why this, and why now

The gap the project was built into

The immediate provocation was a piece of viral content: an unindexed reel captioned *AI is decoding what birds have been saying all along*, composited from three real things — the media cycle around machine learning for animal communication, a photograph of a nightingale, and a Sainburg-style two-dimensional embedding of a vocal repertoire [66]. The composite implies a translator. None of its three ingredients is one.

Every ingredient, however, is buildable, and the honest version of the promise is worth building. Repertoire characterisation works. Functional decoding — linking a signal to a context or a behaviour — works for the handful of systems where playback experiments exist. Recovering propositional content does not work and is not close [65]. There is also a structural asymmetry that any such system runs into: birdsong carries rich syntax and weak or absent reference, while calls carry reference and little syntax [54]. A system that conflates the two produces the reel.

The interesting question is therefore not "can a model translate birdsong" but *what can a large archive of wild recordings actually license*, and what does it cost to say so precisely. That is the question this work answers, mostly in the negative, and the negative answers are specific enough to be useful.

Why a physical model rather than a generative one

Bird-vocalisation synthesis has a validated mechanistic option, which is unusual for a biological signal. The normal-form oscillator reproduces canary song from two scalar gesture curves [24]; the bird's own premotor neurons encode those same gestures [3]; the model runs as a real-time neural prosthesis [4]; and, in the strongest available test, 256 playback trials on 26 wild rufous-collared sparrows at 15 sites found no significant difference between responses to real song and to song synthesised from the model [7]. Heterospecific song evoked no response at all in the same design.

A neural audio generator would have none of that. It also could not carry the evidence labels, because a label has to be grounded in a mechanism a reader can inspect, and a generator's mechanism is a weight matrix. The decision to keep the synthesiser physical is therefore not a tooling preference; it is what makes the honesty rule enforceable.

Why now

Three things changed recently enough to matter. Archive scale: xeno-canto and iNaturalist between them now expose enough recordings that per-species syllable inventories over thousands of species are a matter of compute rather than of fieldwork [10, 75]. Foundation models: Perch, BirdNET, AVES, BirdMAE and NatureLM-audio give an outside opinion on any embedding one builds [29, 35, 59, 64, 73], which changes what a self-reported embedding quality number is worth. And browser capability: a physically integrated syrinx runs in an AudioWorklet at 48 kHz with 16- to 64-fold oversampling, so the interpretable synthesiser and the map can be the same artefact, with no server in the loop.

3 What has already happened

3.1 Physical models of the avian syrinx

The lineage this work builds on treats the syrinx as a low-dimensional nonlinear oscillator rather than as a signal generator to be reverse-engineered. Measurements on the excised syrinx show period doubling, mode locking and chaos, so the source is not a whistle [20]. The minimal dynamical system reproducing that behaviour sits at the Takens–Bogdanov point of the labial model, where a Hopf line is tangent to a saddle-node curve [40, 42]; the book-length account is Mindlin and Laje [46]. Two scalar control curves — air-sac pressure and labial tension — suffice to reproduce canary song [24], and the bifurcation structure carries the timbre: oscillations born in a Hopf are tonal, those born in a saddle-node-on-a-limit-cycle are pulse-like and harmonically dense, and the spectral content index is a function of tension alone [1, 68]. Practical fitting procedures follow: γ from two (f_0, SCI) points [69], $\chi^2(\gamma)$ over segments [55], and a nearest-neighbour lookup in a precomputed table rather than an optimiser [8].

The periphery is a separate and equally well-measured problem. A pure tone in birdsong is a filter effect, not a sinusoidal source: harmonics that appear in a helium–oxygen atmosphere are suppressed in air [5, 51]. The oropharyngeal-esophageal cavity is actively tuned to track the fundamental within a syllable [63], while the trachea is a tube of fixed length that resonates as a stopped quarter-wave pipe at odd harmonics of one body-determined frequency [50, 62]. Beak gape acts as a high shelf on 4–7.5 kHz rather than as a movable formant [50], and it covaries with fundamental frequency in song [52, 56]. Finite-element treatments

of a real tract exist [21, 36]. Above the periphery, respiration supplies the pressure gesture, with mini-breath and pulsatile regimes separated by syllable rate [31] and a two-state air-sac oscillator now published [18]. The two syringeal sides are independently controlled [26, 70], they couple through the shared trachea [41], and two-voice complexity can arise from one side alone [77]. Finally, the model needs noise in the right place: matching the fundamental and the filters was not enough to drive song-selective forebrain neurons, and noise on *tension* — at an intermediate magnitude — was required [2]. Real crystallised song carries about 1 % rendition-to-rendition fundamental-frequency variation [72] with slow within-day drift [74], and syringeal muscle can follow modulation to 250 Hz [15, 61]. A different model class, the two-mass mucosa-wave family, is the right one for doves [16].

3.2 Birdsong syntax

The syntax literature establishes that birdsong is not first-order and that the departures are specific. Bengalese finch song has finite-state structure with repeat-count-dependent transitions, and a repeat-count model is the minimal sufficient description [34, 53]; partially observable Markov models improve on hidden Markov models on the same species [44]. Nightingales hold 180–200 song types with package organisation [33] and countersinging rules [32]. On the reference side, the Japanese tit *Parus minor* combines two call elements into a compound whose effect on receivers is lost when the order is reversed [71], Siberian jay alarm calls encode predator behaviour [27], and the black-capped chickadee's *chick-a-dee* call has four note types in strict order with D-count scaling to threat [22]. Berwick et al. [6] is the standard cautionary reading on how far the analogy to human syntax can be pushed, and argues that what is called birdsong syntax is phonological rather than semantically compositional — which is the reading this work adopts, since it makes no claim above the level of sequence structure. Beyond birds, sperm-whale codas have been factorised into a combinatorial alphabet using structure-first methods [67], which is the template for doing this without playback. And vocal mimics supply the one form of cross-species ground truth available: lyrebird imitations of grey shrike-thrush song elicit responses from shrike-thrushes as strongly as conspecific song does [14], so a mimic and its model are a pair certified perceptually equivalent by the animals concerned.

3.3 Bioacoustic embeddings and audio foundation models

Unsupervised repertoire mapping is an established method. Sainburg et al. [66] project diverse animal vocal repertoires into two dimensions and quantify latent structure across species; Goffinet et al. [25] learn low-dimensional features that resolve individual and group differences. The interactive descendants of this line — Google Creative Lab's *Bird Sounds*, the TensorFlow Embedding Projector, Apple's Embedding Atlas — are where the visual language of the "galaxy" comes from.

The supervised and self-supervised model line has moved faster. BirdNET is an open species classifier trained on the same archives, at roughly a hundred times the scale of this corpus, working at 3-second resolution [35]. Perch, and now Perch 2.0, provides a general bioacoustic embedding over roughly 15 000 classes at 5-second resolution [73]. AVES applies self-supervised speech pretraining to animal sounds [29]; BirdMAE is a masked autoencoder for bird audio [59]; NatureLM-audio is an audio-language model for bioacoustics [64]. Benchmarks and standardised datasets have arrived with them [30, 60]. Two findings from that literature are load-bearing here. First, recording-condition invariance is treated as a training-time problem, via domain-invariant contrastive objectives that pair same-class examples across domains [49]; the closest available prior on the recording channel itself comes from *outside* bioacoustics, where convolving training audio with recorded device impulse responses and mixing frequency-band statistics improves generalisation to unseen recording devices in acoustic scene classification [48]. Second, and against the grain of the first, the AVEX study — an empirical sweep of data diversity, scale, architecture and training recipe over 26 datasets — finds that data diversity dominates: more genuinely different domains per species buys more than the architecture or the loss does [47].

3.4 Methods borrowed from other fields

Four toolkits are used here without modification. Dimensionality reduction and density-based clustering for the map [9, 45]. Continuation-count smoothing for the transition model [37]. Batch-effect correction from genomics and probabilistic linear discriminant analysis from speaker verification, as post-hoc candidates for removing recording-condition structure [38, 57]. Normalised compression distance as a model-free sequence baseline [12], whose underlying normalised information distance is Li et al. [43]. Mel-cepstral distortion as an audible-distance metric [39]. The Menzerath–Altmann law as a falsification test on a unit inventory, which has been applied to gelada and to penguin vocal sequences [19, 28]. Random Fourier features for the browser's kernel ridge [58], a normalised temperature-scaled contrastive loss [11], gradient-reversal adversarial heads [23] and style mixing [76] inside the learned critic. Two methods named as the right next step are simulation-based inference for mechanistic simulators [13] and differentiable digital signal processing [17].

4 Where this work picks up

From the physical-model literature this work takes the oscillator unchanged. The implemented equations are the published normal form term for term, with a sign

convention on x that places the limit cycle at positive pressure and tension; that equivalence was verified numerically before the parameter mapping was fixed. It takes the parameter reading unchanged too: α is air-sac pressure, β is labial tension and sets the fundamental, γ is a pure time-scale. It takes $\gamma = 24\,000$ as the value at which the calibration table is measured, following Perl et al. [55], and the tube formula for the tracheal comb from Nelson et al. [50] and Riede et al. [63]. It takes the placement of noise on tension rather than on pressure or on the output directly from Amador and Mindlin [2], and the 140 Hz corner for it from the respiratory/syringeal modulation boundary [61]. It takes the respiratory oscillator and the two-voice coupling kernel as published [18, 41].

Three departures are deliberate and are the origin of most of what follows.

Per syllable, not per bird. The literature fits γ once per bird. Here every syllable gets its own time-scale, chosen so that the syllable's median pitch sits at $\beta \approx 1$. With γ pinned, the oscillator cannot go below roughly 1.4 kHz, and every owl, dove and cuckoo in the inventory lands one to three octaves sharp because the tension floor binds instead. The consequence, discovered later, is that pinning to $\beta \approx 1$ pins every syllable to the *tonal* end of the model rather than to its middle, which amputates the spectrally rich tail (Section 6).

At archive scale, from measurements only. The literature fits one bird at a time from retained audio. Here the fit runs over 13 859 types across 2 421 species from descriptors and log-mel patches, with the audio deleted. That forces the objective to be a function of what the store kept, which turns out to be the binding constraint on fidelity.

An inventory and a grammar, not a single song. The unit of the system is a per-species syllable-type inventory with a measured transition table over it, so that the map has named regions and the synthesiser can improvise in a species' own measured sequence statistics rather than replaying one fitted phrase.

From the embedding literature this work takes the repertoire-map idea and the finding that recording condition is a first-order confound. It *starts* from the observation that the confound was measured in its own embedding at +0.488 nearest-neighbour lift against species' +0.196, and that this number, not the fit, is what limits every downstream claim.

From the foundation-model literature it takes Perch and BirdNET strictly as external second opinions. Their labels are metadata and are forbidden from becoming typing ground truth: the moment an external inference becomes a type's identity, model error has been laundered into the canonical inventory and the pre-registered-metric discipline stops meaning anything. There is a second reason for that rule which is specific to the resolution mismatch: Perch labels a 5-second window, not a syllable, so background birds can dominate its answer.

5 The system

Figure 2 gives the whole chain. Stages 1–4 run offline in Python; stage 5 is a browser application reading static assets, with no Python process and no model weights.

5.1 Harvest: decode, describe, delete

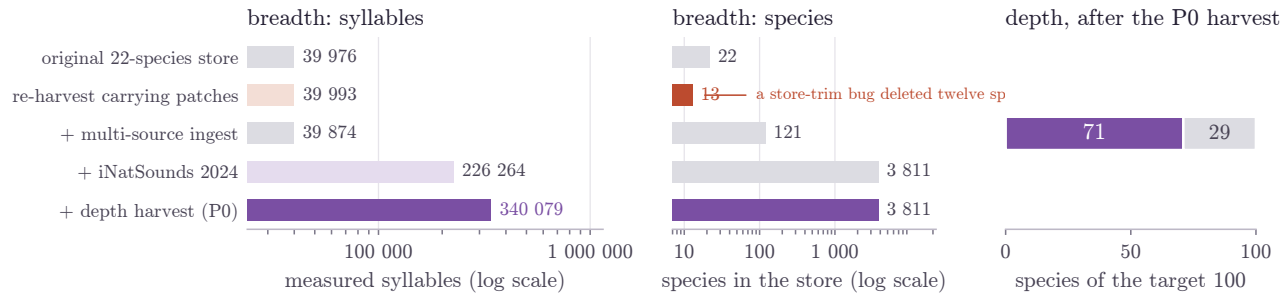
The harvester fetches from xeno-canto and iNaturalist through one loop that decodes a recording, segments it, describes every syllable, and deletes the file; peak memory is one recording. What is retained per syllable is a 28-dimensional descriptor, a 32×23 log-mel patch (736 numbers, log-magnitude only, no phase), a four-knot mel path, a duration, a pitch contour and provenance fields.

The corpus grew in four documented steps: an original 22-species store of 39 976 syllables; a re-harvest carrying patches; a 121-species multi-source ingest; an iNatSounds pass adding 6 453 recordings, 226 264 syllables and 3 811 species in one run; and a depth harvest adding 139 440 syllables and taking recordings from 7 992 to 10 184. The depth harvest targeted 100 species chosen for documented syntax cases, repertoire depth and existing playback anchors, at up to 30 recordings each. It reached a median of 29 recordings per species on those 100, with 71 above the 20-recording threshold the syntax work needs. The tenth percentile is 5, and that is the archive's ceiling rather than the harvest's: xeno-canto holds no more for those birds.

Audio retention was later permitted where the licence allows it, for one measured reason: an external model and any per-syllable re-measurement need the waveform. NoDerivatives is refused, because everything downstream produces derived measurements, and an unrecorded licence is refused, because absence of a licence is not permission. On the production store, 90.3 % of 10 184 recordings pass and all 10 184 carry a recordist. Nothing is shipped to the browser.

5.2 The 28-dimensional descriptor

One descriptor per syllable offline, or per rolling window in the browser, computed sample-for-sample identically on both sides; library defaults are not relied upon. Audio is 48 kHz mono, peak-normalised to 0.9, framed at 1024 with hop 512 under a symmetric Hann window. A frame is voiced if its unwindowed RMS exceeds 10^{-4} and 5 % of the clip maximum. Per voiced frame: 13 mel-frequency cepstral coefficients from 26 triangular filters on the power spectrum with energies floored at 10^{-10} and an unnormalised DCT-II; the spectral centroid as $\log_2$ Hz; and spectral flatness. The descriptor is the mean of the 13 coefficients, their population standard deviations, the mean $\log_2$ centroid and the mean flatness, standardised against corpus statistics shipped with the assets.



Syllables and species are separate panels because they are separate units; the old figure overprinted one on the other. The second row is drawn in clay because a store-trim bug deleted twelve of its 25 species, so any per-species figure measured on it has chance 7.7 % rather than 4.0 % and is not comparable to the rows around it. Right: 71 of the 100 targeted species reach the 20-recording bar the syntax work needs, at a median of 29; the tenth percentile of 5 is the archive's ceiling rather than the harvest's, because xeno-canto holds no more for those birds.

Figure 1: The corpus grew in breadth and then in depth. Syllable counts and species counts are separate panels because they are separate units. The second row is drawn in clay because a store-trim bug deleted twelve of its 25 species, so any per-species figure measured on it has chance 7.7 % rather than 4.0 % and is not comparable to the rows around it. The right panel is the depth harvest: 71 of the 100 targeted species reach the 20-recording bar, and the tenth percentile of 5 is the archive's ceiling rather than the harvest's.

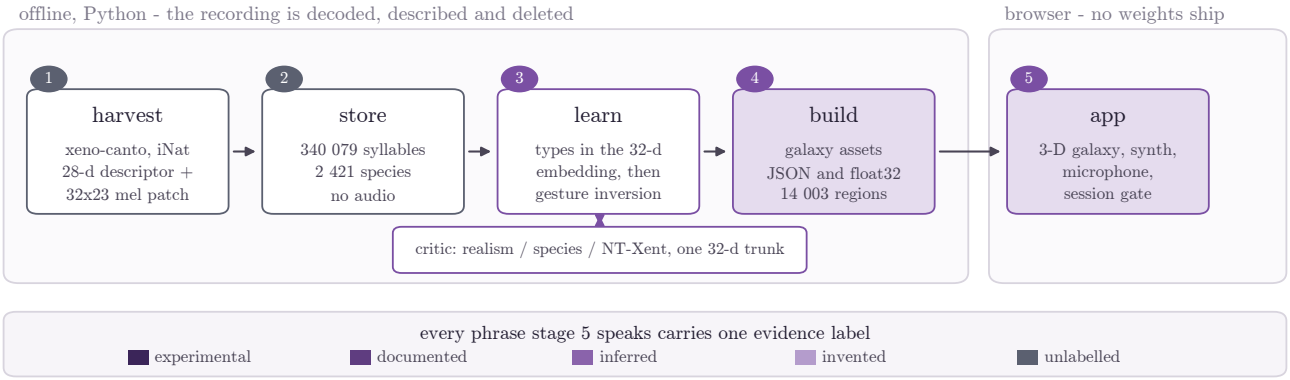


Figure 2: The system end to end. Stages 1–4 are offline and are the subject of Sections 5. Stage 5 reads static JSON and `float32` assets. The evidence labels of Section 1 sit over everything stage 5 displays.

5.3 The critic: one network, three heads

Everything else in the pipeline is hand-written NUMPY. Two quantities, however, are distances rather than physics, and hand-designed distances kept failing. A small convolutional network on the log-mel patch carries three heads off a shared 32-dimensional penultimate layer: a realism logit (real audio against our own resynthesis), a species classifier over the training species, and a contrastive head under a normalised temperature-scaled loss at $\tau = 0.2$ [11]. Species are held out entirely for the realism evaluation, because a network that has seen every species can report identity instead of realism. Augmentations are small time stretches, gain changes and added noise — never pitch shifts, since pitch is type-diagnostic here.

Forcing all three tasks through one 32-dimensional bottleneck is the point: the embedding has to answer all three at once. The production critic trained on 2 686 species with 895 held out reaches realism AUC 1.000 on held-out species, species accuracy 0.730 over 2 686 classes against chance 0.0004, and novelty ROC 0.846 flagging 32.2 % of unheard species' syllables at a chosen 5 % false-positive rate.

The network never generates. It scores, embeds and flags novelty; what ships is an interpretable parameter vector rendered by the physical model.

5.4 Typing, and the fit

Syllable types are k-means clusters in the 32-dimensional embedding, with k chosen per species by silhouette. Replacing the previous hand-weighted seven-number observation vector with the embedding is the largest single accuracy gain in the project (Section 7).

Each type is then fitted to the oscillator. The cluster median supplies a measured pitch contour, duration, timbre class, attack and modulation depths. Coordinate descent (`invert_gestures`) then searches thirteen gesture axes — including `gammaScale`, `tractQ`, `noise` and `rough` — over three passes, a few hundred renders per type, moving only on an improvement. Measured contour and duration are excluded from the search deliberately: the pitch tracker reaches them to about 1 %, and a search should not overwrite a measurement.

The objective is a cosine between the resyn-

thesis's 28-dimensional descriptor and the type's mean-member reference, in the standardised harvest population frame. That single scalar decides the gesture search, the respiration and two-voice ship gates, and the fricative routing. Section 6 reports that it does not rank audible fidelity.

Two per-species fits sit beside the per-type one. The tracheal comb's fundamental f_1 is *measured*, not searched: per-syllable cepstral means are averaged over all of a species' observations, inverted to a 26-band log spectral envelope, detrended, and scanned for the f_1 whose odd-harmonic series best aligns with the residual. Because those syllables span many recordings and a wide pitch range, pitch-dependent structure averages away and only resonances fixed in hertz survive — the long-term-average-spectrum argument applied to a store that kept no audio. The measurement proposes and the resynthesis decides: a species whose recordings show no tracheal colour keeps none. Implied effective tube lengths came out at 1.7–6.6 cm against 3.4–4.5 cm measured in the literature.

5.5 Grammar and songs

A gap longer than 1.0 s breaks a bout; that constant is shared by the grammar, the chunk miner, the bake-off and the song detector through one named parameter, after a period in which it existed as three separate literals. The shipped grammar is a first-order transition table with Kneser–Ney continuation-count backoff [37], plus 26 897 measured per-transition gaps, so nothing about the rhythm is invented.

Song types are detected rather than authored. Each bout's repeat runs are collapsed, the collapsed strings are clustered by normalised edit distance, and a cluster of at least six occurrences becomes a *motif* song type. A second class was added after the funnel was counted: a *repeat* class is one collapsed string heard at least six times in bouts of at least three syllables, matched by equality, because equality is the only similarity a one-token string carries honestly. Both classes ship with a `kind` field so nothing downstream has to infer which claim is being made. Every configuration is judged against a within-species permutation control that keeps bout lengths and token marginals.

5.6 The map and the live path

The build projects the 32-dimensional embedding to three dimensions, clusters it, and writes plain JSON and `float32` assets: one position per syllable, per-cluster centroids with labels, functions, confidences and citations, and the cross-species relations document. Three clustering modes exist. The default is seeded k-means, where each type contributes a seed and the resulting partition votes a name back by point-count majority. HDBSCAN is the discovery mode, and is the only one that can answer "none of these". Label assignment sends every point that already carries a type to its own type's cluster and fits k-means only over the unlabelled remainder; Section 7 reports what that is and is not worth.

The browser cannot run the critic — no weights ship, which is a hard constraint rather than a packaging detail — so the live path needs a bridge from the 28-dimensional descriptor it can compute to the 32-dimensional space the map lives in. That bridge is a kernel ridge over random Fourier features ($D = 2048$, $\sigma = 4$, $\lambda = 1$), fitted offline and shipped as plain numbers [58]. It is gated on held-out *recordings*, never on held-out syllables, because two syllables of one bird in one take are nearly the same pair twice. A linear ridge came first and was replaced by measurement, not by preference: it cleared the 0.8 gate at 0.8585 on a 19-species critic and fell to 0.7997 once the embedding grew finer.

Live activation is a cosine against every centroid, a softmax at a shipped temperature, and an exponential moving average at 0.82/0.18, which is about 120 ms of smoothing at 60 Hz. A per-point brightness term makes a cluster light from the inside out — the members nearest the live sound brighten first — so an active region shows its internal structure instead of switching on as one blob.

5.7 Verification

The two renderers must agree sample for sample. A parity harness renders a fixture list through the real AudioWorklet and through the Python renderer and compares them; it stands at 18 of 18 fixtures with the worst divergence at float32 rounding (2.8×10^{-8}). Six divergences had accumulated before it existed, including amplitude modulation applied twice in the browser and a per-integration-step clamp that flattened the high-amplitude regimes. A separate check renders all 44 authored syllables and measures pitch, currently at 0.3 % median error. Forward Euler is validated against Runge–Kutta to within 2 % of f_0, and the calibration table reproduces the reference oscillator measurements to 0.1 % on f_0 and 2 % on peak-to-peak amplitude. The noise generator is part of the contract: xorshift32 with a specified seeding rule, so that "the same turbulence" is a claim two implementations can be tested against.

6 What is done differently, and where it stops

6.1 The contribution, stated plainly

Four things here are not standard practice in this area.

A physical inventory at archive scale. 13 859 syllable types across 2 421 species, each one an interpretable parameter vector for a published oscillator rather than a latent code. To our knowledge no comparable inventory exists; the physical-model literature fits individual birds.

A corpus that is a description rather than a copy. The store holds no audio and cannot be inverted to recognisable audio, which changes what can be shared and removes the licensing question from the analysis entirely.

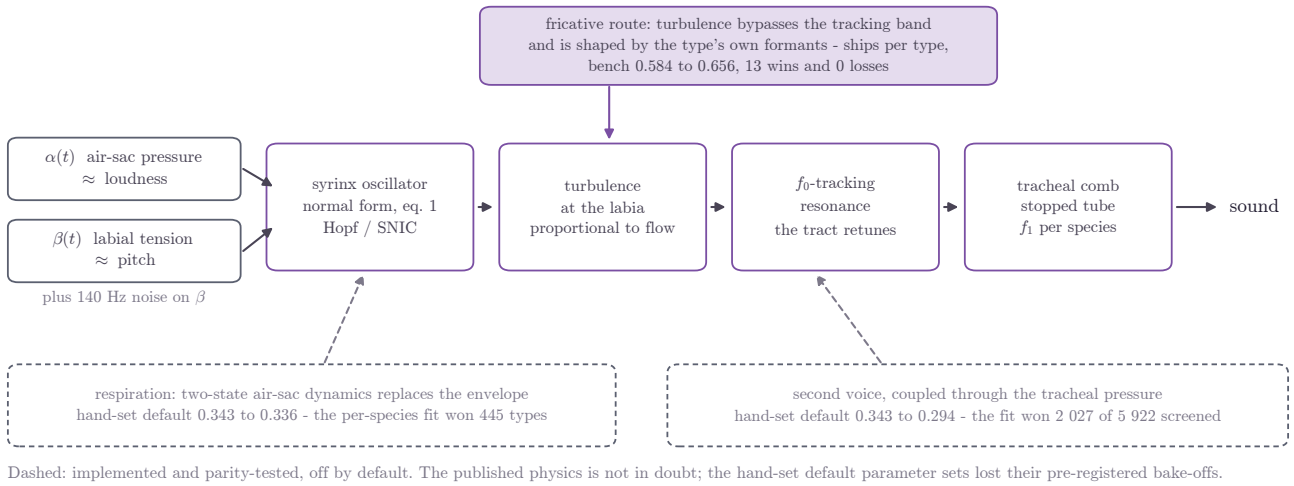


Figure 3: The synthesis chain. Two gesture contours drive the oscillator; turbulence, the f_0-tracking resonance and the fixed tracheal comb shape the result. The dashed blocks are implemented, parity-tested and off by default: their published physics is not in doubt, but their hand-set default parameter sets lost their pre-registered bake-offs, and only the per-species fits ship, per type, where they win.

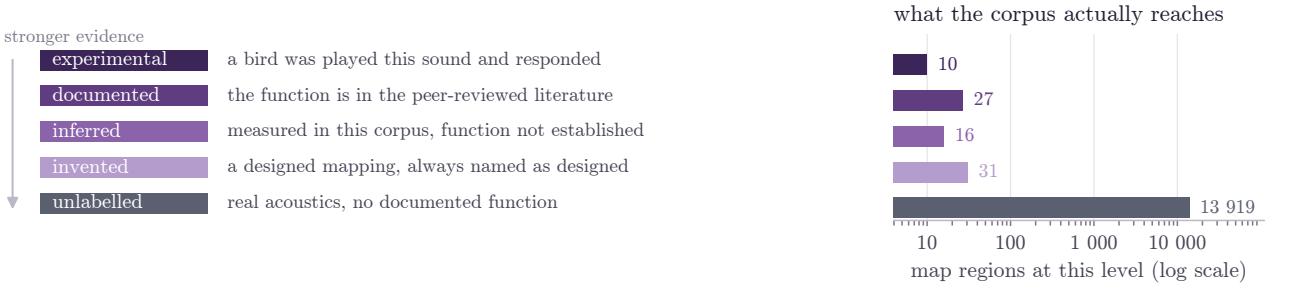


Figure 4: The five evidence labels, the requirement for each, and the corpus counts at the time of writing. Every phrase the application speaks carries one of them; nothing renders without one.

Evidence labelling as an enforced invariant. Five levels, applied to every rendered phrase and every named region, with the "unlabelled" level protected: detecting that a sound is new is not knowing what it means, and the interface is required to keep saying so.

Pre-registration as the working method. Fifty-eight gates, written before their runs, with the refusals kept in place. This is why the rest of this paper can be specific about what the data does not support.

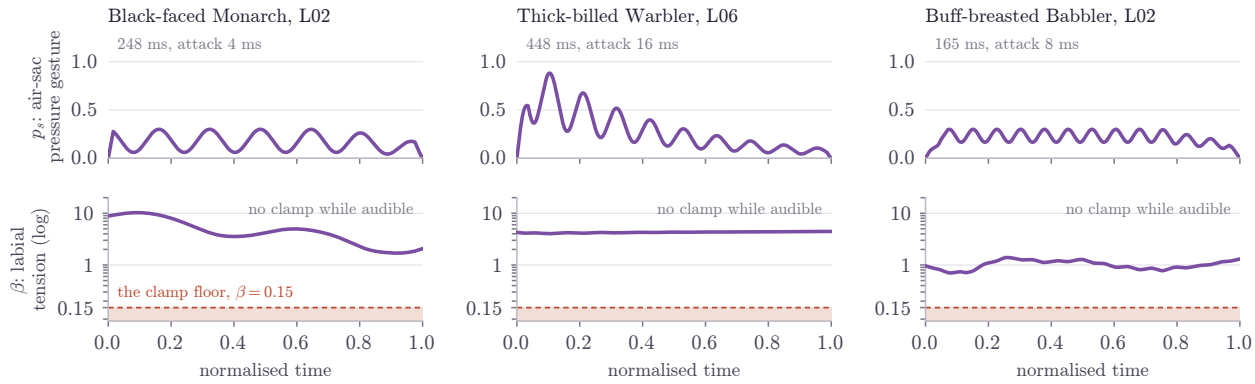
6.2 Wall 1: the objective cannot hear

The pipeline's universal fit objective is a 28-dimensional descriptor cosine. Measured against two independent audible-distance metrics on 200 fitted types — mel-cepstral distortion [39] and multi-resolution log-spectral distance — it carries essentially no information: Pearson +0.106 and Spearman +0.100 against MCD, Pearson +0.102 and Spearman +0.111 against STFT error. Neither is distinguishable from zero and both point estimates lean the wrong way. This is not an instrument artefact: the recomputed cosine reproduces the stored one at Pearson 0.921, and the two distortion metrics agree with each other at 0.925.

The mechanism is legible. High-cosine types are clean whistles whose renders concentrate energy in one band while the field audio carries broadband noise everywhere else, so the spectral distance is large; noisy buzzes score a low cosine but have broadband energy on both sides, so the distance is small.

The obvious replacement was tried and does not work as an optimiser. A time-resolved mel-path target correlates with MCD in the right direction and at usable strength (Pearson −0.305, Spearman −0.420), which is what a fidelity ruler should do. Blended into the objective at weight 0.5 it wins 31 % of types on MCD against a pre-registered 60 % bar, with a median ΔMCD of exactly 0.0 dB and a maximum cosine loss of 0.120. A ten-type probe had won 7 of 10; the 80-type gate run settled it.

The honest state is therefore that the project measures spectrally and optimises descriptor-wise. Every cosine-gated adoption in the system inherits that caveat, which is recorded rather than hidden: the respiration and two-voice fit winners, the fricative routing and the one-tone texture re-fit were all decided by a proxy that does not rank audible fit at type level.



The two gestures the oscillator is driven by, for the same three types as the acoustic examples, read straight out of the shipped fit through the repository's own render planner and its measured tension table. This is what interpretable parameters buys: every sound in the corpus is two named physical contours and thirteen searched gesture axes, not a latent code. Pressure and tension are separate panels because they are separate units. The clamp annotation is measured per type, weighted by where the envelope is actually open; across the production learn the clamp bites in 6 617 of 12 724 types.

Figure 5: The fitted gestures behind the three acoustic examples. Every syllable in the corpus is two named physical contours — air-sac pressure and labial tension — rather than a latent code, and this is the whole of the interpretability claim. The dashed line is the tension clamp of the parameterisation; a contour that rides it is at a limit of the parameterisation rather than at a fitted value, which happens in 6 617 of 12 724 types across the production learn.

They are not necessarily wrong — each also improved the descriptor match — but "improved the proxy" is a weaker claim than it reads.

6.3 Wall 2: the recordist, not the bird

In the critic's embedding, the strongest relation is not the animal. Nearest-neighbour lift over a permutation null, blocked by recording, is +0.488 for recordist and +0.196 for species. Every consumer of the embedding inherits that: typing, the map, the relations document, novelty and the unknown-bird path.

The road out has now been closed at four independent ends, which is Section 7's and Section 8's subject and is summarised here. Training-time augmentation that simulates a recorder's frequency response moves the recordist lift by −0.7 %, i.e. not at all. A gradient-reversed recordist head [23] does move it, at an almost exactly one-for-one cost in species structure. A species-factored version of that head protects species completely and reaches only −6 % of the confound. Post-hoc batch correction, including the strongest member of the genomics toolkit [38], reproduces the same one-for-one exchange rate by a completely unrelated mechanism. And two archive-scale external models, neither of which has ever been asked to tell our resynthesis from a bird, still carry between five sixths and more than all of the recordist structure that ours does.

The reason is now visible and is a statement about the sample rather than about any model: most species in this corpus are carried by a handful of recordists, so no method that sees only the embedding can separate labels the data never separated. Subtracting each recordist's mean leaves the recordist lift *higher* than it started (0.1665 to 0.1687) while costing a third of the species structure, so the confound is not a location shift and not a per-dimension gain either. The named action is more recordists per species, in the data.

6.4 Wall 3: syntax is data-limited, not method-limited

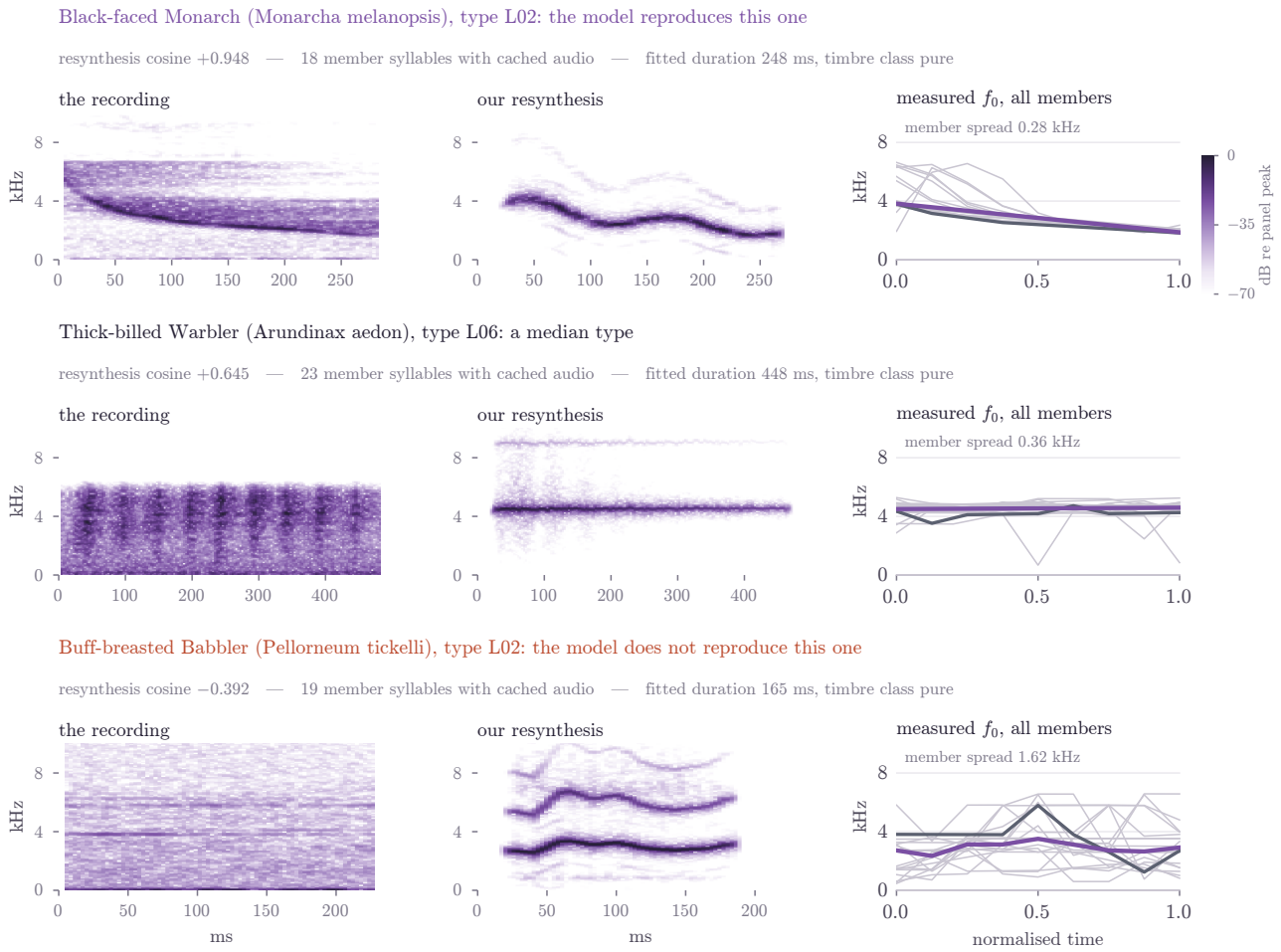
A first-order transition model over the learned tokens is strongly supported (Section 9). A second order is not, and the reason is countable: of 2 421 species, 217 clear the 20-bout bar at which the question can be asked at all. Within those, the median species *loses* 0.091 bits per token going from order 1 to order 2 on bouts it has not seen, at three independent splits. The literature is unambiguous that birdsong is not first-order [34, 44], so this is a statement about the corpus, not about birds. The depth harvest is the action, and it has a hard ceiling: the tenth percentile of the 100 deepest species is 5 recordings, which is what the archive holds.

6.5 Wall 4: the map's own confusion

Two ceilings sit on the claim "sing, and the right region lights".

The first is in the embedding. With every type granted its own cluster — so that no type can fail to be retrieved for lack of a cluster — the share of labelled points whose nearest centroid by cosine belongs to their own type is 0.602. For 40 % of points the nearest centroid is another type's. That is the embedding's confusion between types, not the clustering's, and it was invisible until label assignment removed the competing explanation.

The second is in the bridge. On held-out recordings, the share of syllables whose nearest region is the same through the ridge map as through the critic itself is 0.61 on the debug build and 0.66 on production. Decomposing 1 439 flips over 3 731 held-out syllables locates it: 76.4 % is boundary near-ties, with 74 % of all rows sitting inside a margin of 0.05 and 41 % of flips landing on the runner-up region. Genuine large-margin map failure is 4.3 % of the loss. Tie-adjusted agreement is 0.91 and species-level agreement is 0.781



Each spectrogram: 22.05 kHz cached audio, 256-sample Hann window, 64-sample hop, peak-normalised, 70 dB range. Both sides of a row carry identical settings, identical axes and one colour scale. The resynthesis is rendered by the repository's own renderer at 48 kHz and resampled to the cache's rate; 20 ms of context is kept on the recording and matched with silence on the resynthesis, so the flat band at each panel's edge is the field recording's own noise floor. Third panel, grey: the measured pitch contour of every cached member of the type; slate: the member plotted to its left; violet: the contour the fit ships. The cosine is the objective the fit was run under; docs/19 measured its correlation with audible distortion at about +0.1, so it ranks the fit's own criterion and not audible fidelity. Production store, and the audio cache holds iNaturalist recordings only.

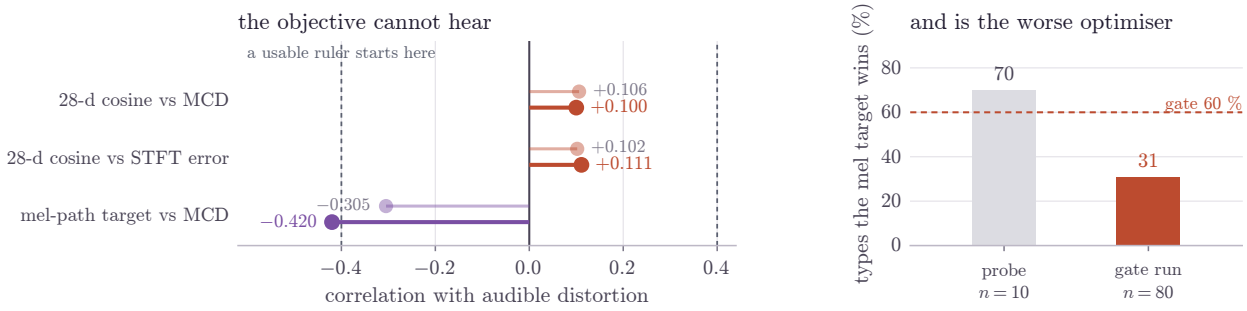
Figure 6: What the physical model reproduces, and what it misses. Three fitted types, taken from the top, the middle and the bottom of the resynthesis-cosine ordering over the types whose members have cached audio; the corpus median is 0.648 over 13 859 types, so the middle row is a median type rather than a chosen success. Top: a clean descending whistle, which the oscillator and its tracking tract reproduce closely. Middle: the median case — the pitch is right, and the pulse train that carries the real syllable's timbre is replaced by a steady tone. Bottom: a failure, and the third panel says why — the type's member contours disagree by kilohertz, so the cluster median the fit starts from describes no member well. The cosine is the objective the fit optimised and is *not* a ranking of audible fidelity (Section 6).

against chance 0.033. The consequence adopted was to make the honest single-label claim at species level and to label region-level claims as "nearest of several close regions"; no better map class was pursued, because the addressable loss is at most 2.6 % of rows.

6.6 Wall 5: the fit, not the model

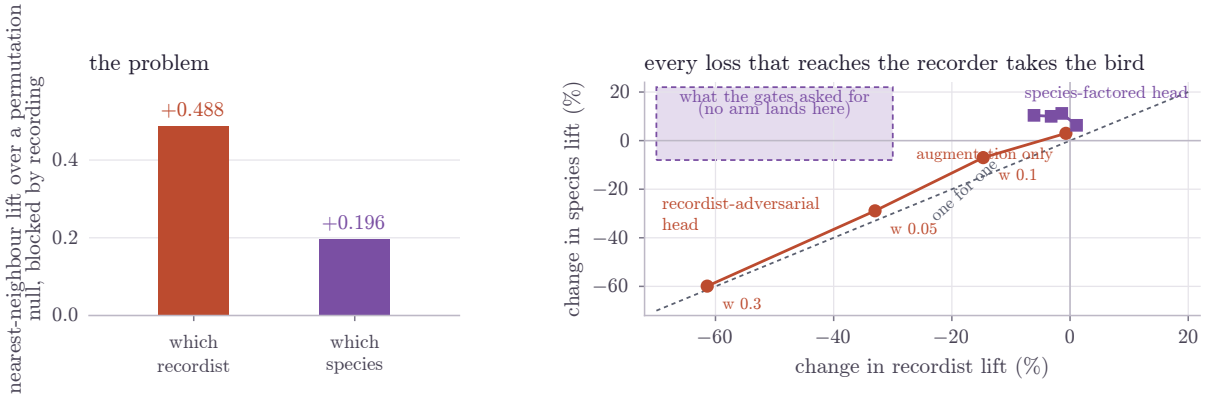
An audit against the canonical literature found no deviation inside the oscillator equations. The implemented ODE is the published normal form term for term, validated against Runge–Kutta, against the reference oscillator measurements, against a two-sided parity harness, and against the literature's own stated coupling-approximation boundary — the two-voice kernel's measured collapse into period doubling occurs exactly where Laje and Mindlin [41] says the weak-coupling approximation dies.

Every recorded deviation sits outside the oscillator, and the list is specific. A single f_0-tracking bandpass makes the spectral envelope pitch-invariant: measured relative partial gains span 1.3-fold over a 5.7-fold change in f_0, against 32-fold for a fixed waveguide, so the structure locks spectral shape to pitch. Pinning every syllable to $\beta \approx 1$ places it at the tonal end of the model: the resynthesis reproduces the median spectral content index almost exactly (1.099 against 1.092) and has essentially no rich tail (1.4 % above 1.5 against 15.6 % of real syllables; 0 % above 3 against 5.3 %), with median spectral flatness three orders of magnitude too pure at 0.0005 against 0.044. The tension clamp bites in 6 617 of 12 724 types, so over half the corpus has contour excursions riding a parameterisation limit while audible. Pressure arrives as a designed envelope rather than as respiratory physics. And there is no physiological micro-variation: real crystallised song



Left: pale and thin, Pearson; solid and thick, Spearman. Neither of the cosine's correlations is distinguishable from zero and both point estimates lean the wrong way. Not an instrument artefact: the recomputed cosine reproduces the stored one at Pearson 0.921 and the two distortion metrics agree with each other at 0.925. Right: blended into the objective at weight 0.5 the mel target wins 25 of 80 types, with a median ΔMCD of exactly 0.0 dB and a maximum cosine loss of 0.120; seven of ten had won at probe scale. $n = 200$ types then $n = 80$; both panels are debug-scale.

Figure 7: Left: correlation with audible distortion for the incumbent objective and its proposed replacement, $n = 200$ types on the 100-species debug store against the cached audio. Right: the replacement's win share at exploratory scale and at pre-registered gate scale. Both panels are debug-scale.



Left: in the critic's embedding the strongest relation is not the animal, and every consumer inherits it - typing, the map, the relations document, novelty, and the unknown-bird path. Production store. Right: every swept point of both adversarial heads. The shaded region is what the pre-registered gates asked for and no arm reaches it; the diagonal is a one-for-one exchange of recordist structure for species structure, and the plain head tracks it. The species-factored head protects species completely and reaches only 6 % of the confound. 37-species debug store; debug-scale.

Figure 8: Left: what the embedding organises by. Right: every swept point from both adversarial heads, as relative change in recordist lift against relative change in species lift. The shaded region is what the pre-registered gates asked for and no arm reaches it. 37-species debug store; debug-scale.

carries about 1 % rendition-to-rendition fundamental-frequency variation [72], and the implemented roughness mechanism is one static fitted depth per type.

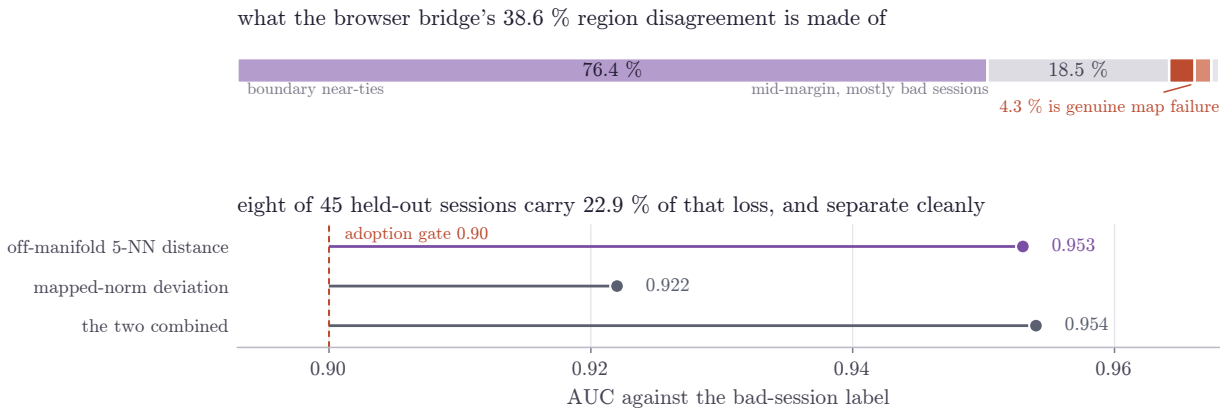
Two reading notes follow. First, "noisy" and "machine-like" are opposite failure modes of one cause family: the representation has one noise axis and one envelope, so a fricative collapses to unshaped hiss and a tonal note to a metronomic sine. Real syringes live between the regimes, and the model contains that middle — period-doubled and chaotic solutions are in the equations [1, 68] — but the fit cannot find its way there under the current objective. Second, the diagnosis is now sharper than "the fit fails", because the starting point has been ruled out: an amortised inverse encoder that proposes a full parameter vector in one forward pass reaches a comparable basin about as often as the cluster median already does (Section 7). What remains is the objective and the physics beyond the oscillator.

7 Our methods against each other

This section reports the head-to-head arms the project ran against itself. Each states what it licenses and what it does not. Unless noted, arms sharing a table were run on one fixed corpus or one fixed dumped matrix, through the shipped functions rather than through re-implementations, so they are comparable to each other and not necessarily to arms in other tables. Throughout, "points" means percentage points: a retrieval score of 0.5775 rising to 0.6020 is +2.5 points.

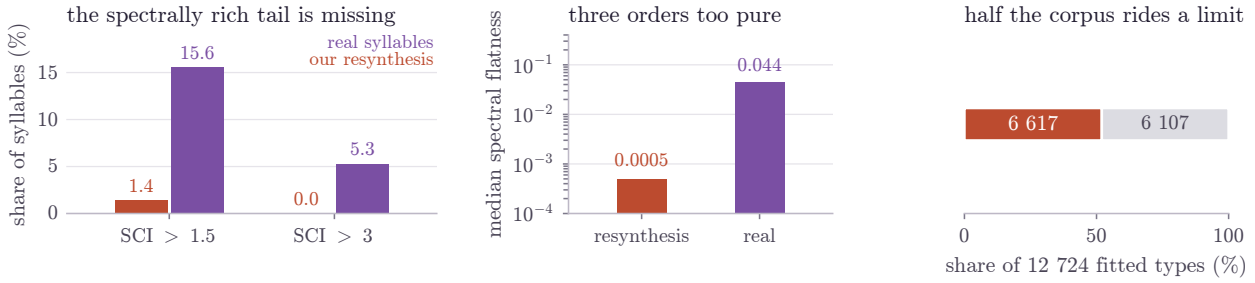
7.1 Hand-weighted typing against the learned embedding

Syllable identity was originally decided by seven hand-weighted numbers. Replacing that vector with the critic's 32-dimensional embedding is the largest accuracy gain the project has measured. The headline metric is species placement: does a resynthesis land nearest



Top: 1 439 flips over 3 731 held-out syllables. Tie-adjusted agreement is 0.91 and species-level agreement, which is the honest single-label claim, is 0.781 against chance 0.033; the addressable loss is at most 2.6 % of rows, so no better map class was pursued. Bottom: the browser can compute the winning signal from the 28-d side alone, and a shipped cloud of 256 random anchors holds AUC above 0.92 for 28 kB. Using it as a lever - raising the remaining rows' agreement by 5 points - failed at 3.5, so the pre-registered fallback fired and it ships as an honesty gate. Debug build.

Figure 9: Top: the 38.6 % region disagreement between the browser's bridge and the critic, decomposed over 3 731 held-out syllables. Bottom: separation of the eight pathological sessions by each candidate live signal, against the pre-registered adoption gate. End-to-end debug build; debug-scale.



Pinning every syllable to $\beta \approx 1$ places it at the tonal end of the model. The median spectral content index is right - 1.099 against 1.092 real - and the tail is not there at all: the model contains the pulse-like regime that produces it and the fit does not reach it. Median spectral flatness is three orders of magnitude too pure, which is a whistle too clean to be an animal. Right: the tension clamp bites in 6 617 of 12 724 types while the syllable is audible, so over half the corpus has contour excursions riding a parameterisation limit rather than a fitted value.

Figure 10: The tonal-regime pin. Left: share of syllables above two spectral-content thresholds, resynthesis against real. Right: median spectral flatness on a log axis. The model contains the pulse-like regime that produces the missing tail; the fit does not reach it.

its own bird's centroid rather than another species'?

Table 1 licenses three statements and refuses a fourth.

Learned typing is a clean win, replicated on two stores at two species counts: +6.3 points at 9 k and +13.3 points at 40 k, with the guardrail cosine essentially held and pitch error improving in both. Nothing regresses.

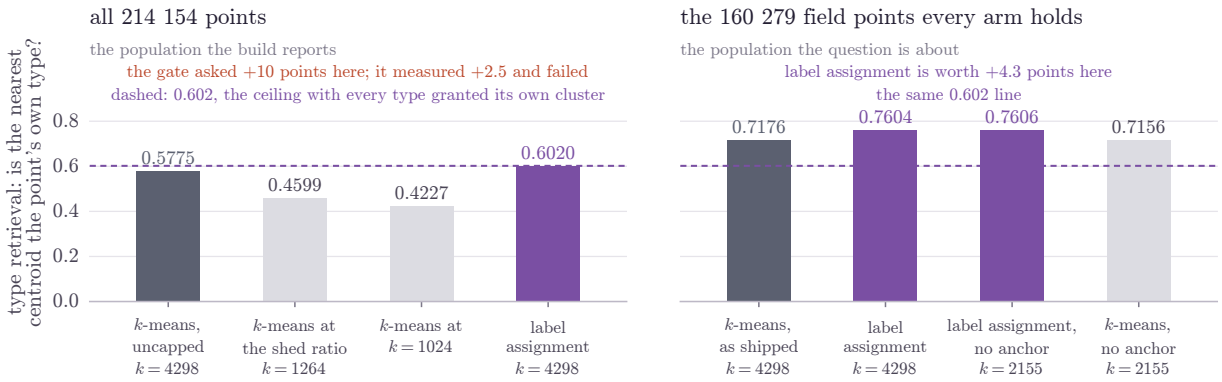
The learned realism logit as a *fit objective* is refuted, and it fails in exactly the predicted way: monotonically harmful in its weight, with the cosine collapsing from 0.762 to 0.335 at $w = 1$ while placement drops to 17.3 %. A critic at realism AUC 0.990 on entirely held-out species rules out "the critic was not good enough". Separating two distributions is not the same as having a gradient that points at the nearer one. The guardrail term earned its place by catching the thing it was put there to catch, and the default weight is zero.

The own-type placement column, which we omit from the table, moves the wrong way as species placement rises, and that is an artefact rather than a regression: the learned embedding drove the type count from 138 to 294, and own-type placement asks whether a resynthesis beats *every other type of the same bird*. With twice as many rivals it is a strictly harder question, and it is not comparable across rows with different type counts. That is why species placement is the headline.

What the table does *not* license is the waveguide row. A structurally fixed tract with 32 times the spectral freedom of the tracking band improves all three metrics at once on five species and was a regression across 22 species on a different store; here it wins under both typings and pays for it in cosine, missing the pre-registered guardrail. It remains promising and unconfirmed, and it is gated off.

7.2 Two spaces: the 28-dimensional descriptor against the 32-dimensional embedding

The two spaces are not interchangeable and the choice is measured, not assumed. On a controlled sub-



The question is the one the browser asks of live audio: is the nearest centroid by cosine the point's own type? One fixed matrix dumped from a 100-species debug build - 214 154 points, 2 143 types, 160 279 of them from field recordings - with every arm scored through the shipped clustering, seeding and purity functions. On a corpus that fits under the cluster cap, seeded k-means is nearly as good as knowing the answer, which is why the gate fails and why the failure is the useful part. Changing the query population moves the same arm by 14 to 16 points, more than any arm moves it, and an arm that drops the synthetic copies drops the hard queries, so on all points it flatters itself by 18. Purity and coverage are identities under label assignment, not measurements. Debug-scale.

Figure 11: Type retrieval against the cluster budget, and against the query population. The question is the one the browser asks of live audio: is the nearest centroid by cosine the point's own type? Left: scored over all points, label assignment beats the shipped k-means by 2.5 points against a pre-registered 10, and fails — on a corpus that fits under the cluster cap, seeded k-means is nearly as good as knowing the answer. Right: scored on the field points every arm holds unchanged, which is the population the question is about. The query population changes the same arm's answer by more than any arm changes it. Debug-scale.

Table 1: Typing space and fit objective, ablated on two stores. Each row changes one thing. *Species* is the share of resyntheses landing nearest their own species; *cosine* is the guardrail the critic cannot touch. Absolute values are not comparable between the two panels: the stores differ, and the right-hand panel is a 13-species measurement after a store-trim bug deleted twelve species, so chance there is 7.7 % rather than 4.0 %.

	9 041 syllables, 25 species				39 993 syllables, 13 species			
arm	types	species	cosine	$\lvert f_0 \rvert$	types	species	cosine	$\lvert f_0 \rvert$
hand typing, cosine objective	346	26.6 %	0.762	5.4 %	138	38.4 %	0.807	3.3 %
+ realism objective, $w = 0.25$	346	27.2 %	0.686	6.6 %	—	—	—	—
+ realism objective, $w = 0.50$	346	23.7 %	0.655	6.5 %	138	31.9 %	0.642	4.0 %
+ realism objective, $w = 1.00$	346	17.3 %	0.335	5.7 %	—	—	—	—
learned-embedding typing	319	**32.9 %**	0.755	3.1 %	294	**51.7 %**	0.796	2.2 %
hand typing + waveguide tract	346	30.1 %	0.719	5.4 %	—	—	—	—
learned typing + waveguide tract	319	37.0 %	0.706	3.6 %	294	57.1 %	0.723	2.1 %

set — 625 field syllables, three species, identical points, one seed, only the space changed — moving the build into the embedding takes cluster purity from 70.9 % to 90.9 %, type coverage from 12/14 to 14/14, three-dimensional 10-nearest-neighbour agreement from 64.5 % to 83.2 % and distance faithfulness from 0.874 to 0.940. On the 100-species debug corpus the descriptor-space discovery map degenerated to three clusters with one holding 95 % of points — iNaturalist soundscape audio is one density blob in 28 dimensions — where the embedding gives 98 clusters with real structure.

Two honest counterweights sit against that. A render-free placement check scores the 32-dimensional space slightly *below* the 28-dimensional one (38.5 % against 41.3 %), and the recordist confound of Section 6 lives in the embedding, not in the descriptor. The embedding is the better similarity space and the worse-behaved one.

7.3 Objectives: cosine against a mel-path target

Reported above in Section 6 and summarised here for the comparison table. The mel target is the better *measurement* and the worse *optimiser*: it ranks audible distortion where the cosine cannot (Spearman −0.420 against +0.100), and as an objective at $w = 0.5$ it wins 31 % of 80 types against a 60 % bar with a median effect of exactly zero. The two objectives agree only moderately on which types are good (Pearson 0.456), so they are complementary rather than redundant. An objective that is both a good ruler and a good optimiser remains unfound.

7.4 Clustering: seeded k-means against label assignment

Ninety-six per cent of a production corpus's points already carry a *(species, syllable)* label by construction. The default build discards it, partitions the space geometrically, and asks each partition to vote a name back



Left: 625 field syllables, three species, identical points, one seed. Hollow marker, the 28-d descriptor; filled, the 32-d embedding. Clay is the counterweight, because render-free placement moves the wrong way and the recordist confound lives in the embedding rather than in the descriptor: the embedding is the better similarity space and the worse-behaved one. Right: the browser computes only the 28-d descriptor, so a kernel ridge over random Fourier features ($D = 2048$, $\sigma = 4$, $\lambda = 1$) maps it into the 32-d space the map lives in. It is gated on held-out recordings, never on held-out syllables, because two syllables of one bird in one take are nearly the same pair twice; it ships as plain numbers, and no weights reach the browser. Both panels are debug-scale.

Figure 12: The two feature spaces and the bridge between them. The browser computes only the 28-dimensional descriptor; the map lives in the critic's 32-dimensional embedding; a kernel ridge over random Fourier features maps one into the other behind a ship gate on held-out recordings.

by point-count majority. Label assignment takes the other side of that design decision: every labelled point goes to its own type's cluster and k-means fits only the unlabelled remainder.

All arms in Table 2 run on one fixed matrix — 214 154 points, 2 143 types, 160 279 of the points from field recordings — dumped from a 100-species debug build, and every arm is scored through the shipped clustering, seeding and purity functions. The harness was validated twice before any arm was read: the $k = 1024$ row reproduces the earlier baseline exactly, and the $k = 4298$ row reproduces the shipped debug build's own manifest.

The pre-registered gate asked for +10 points of type retrieval against the shipped k-means build. It measured +2.5 and **failed**. The failure is the useful part. The mechanism the gate was predicted from is types shedding their only cluster, and under the raised cluster cap this corpus wants 4 298 clusters and gets all 4 298, so that mechanism never fires. *On a corpus that fits under the cap, seeded k-means is nearly as good as knowing the answer* — a real result about the seeding, and the argument against making label assignment the default. Placing this corpus in production's shedding ratio ($k = 1264$) and at $k = 1024$ gives +14.2 and +17.9 points. The scope of the claim is therefore exact: label assignment is worth what the cluster cap costs, and nothing more.

Two of the build's headline numbers stop being measurements under this arm. Purity 0.9988 and type coverage 1.0000 hold because a type *is* a cluster here, by construction. The shipped manifest says so in a dedicated field, because 1.0 read as a good map is reading the definition back. Type retrieval is the statistic that survives, because nothing forces it.

There is a measured cost, and it is in the live path. Against the k-means build of the same corpus, region-level agreement through the bridge falls from 0.6572 to 0.6316 and species-level agreement from 0.7421 to 0.6831, while the tie-adjusted figure *rises* from 0.8032 to 0.8136. Read together that is more near-ties rather than worse decisions: centroids placed at label means are less separated than centroids Lloyd's iteration pushed apart, so more live sounds land between two regions — and the ones that land decisively land better. A build whose headline is the species claim should read that row first.

7.5 The synthetic member anchor ("W5")

The map was built out of both real and synthetic points: each type contributed one rendered anchor plus renders of up to 24 measured member realisations, alongside the field syllables those members already are. At production scale that made 51.1 % of the map a rendered copy of the other half (347 531 synthetic against 340 079 field). Because the critic's leading loss term is realism, the recorded/synthesised axis is the strongest relation in its embedding (lift 0.990 against species' 0.513), so every type lived in two blobs and the cluster budget bound at twice the type count.

Three arms were run on the fixed matrix. Reducing member variants to the anchor alone moved seeds per type from 2.000 to 1.993, i.e. not at all — the source split is *binary*, splitting whenever a type holds points on both sides, so 24 renders and one render split it equally. Deflating the source axis worked exactly as specified, removing 42.0 % of the space's energy and driving the gap between the two source means from 7.233 to 0.0, and *every type still split* (1.999), with purity falling 1.1 points. So the recorded/synthesised difference is not one direction: the global offset is, and the per-type geometry the seeding gates on is not. What works is rendering nothing at all for a type that has member observations, since it already exists in the map as the real syllables the typing assigned to it: seeds per type falls to exactly 1.000 and the wanted cluster count halves from 4 298 to 2 155.

The follow-up changed what that arm is *for*, and it is the cleanest methodological lesson in this section. Type retrieval over *all* points is not comparable

Table 2: Clustering arms on one fixed 100-species debug matrix (214 154 points, 2 143 types). *Type retrieval* asks, for each point, whether the nearest centroid by cosine — the lookup the browser performs on live audio — belongs to that point's own type. Purity and coverage are identities under label assignment, not measurements. Debug-scale.

arm	k	purity	type coverage	type retrieval	clustering (s)
k-means, uncapped (as shipped)	4 298	0.5797	0.9258	0.5775	64.9
k-means at production's shed ratio	1 264	0.4601	0.5114	0.4599	25.5
k-means at $k = 1024$	1 024	0.4232	0.4260	0.4227	27.2
label assignment	4 298	*0.9988*	*1.0000*	**0.6020**	**0.2**

between arms that hold different points: dropping the synthetic copies drops the hard queries, so the arm flatters itself by 18 points. Scored instead on the 160 279 field points every arm holds unchanged — which is the population the question is about, since a recorded syllable must light its own region — the picture is different.

The anchor was removed on the size argument, not the accuracy argument, and the decision is recorded as such. It halves the region count and removes a quarter of the points, taking the production build from a wanted 27 862 clusters to 14 003 and the centroid asset from a projected 44 MB to 7.69 MB of JSON plus 3.81 MB of binary sidecar — the difference between "needs a new asset format" and "ships". It costs the rendered voice its own place on the map, which is a claim about our synthesiser rather than about a bird. One number moved the wrong way and it is a picture number: three-dimensional distance faithfulness fell from 0.903 to 0.747 while 10-nearest-neighbour agreement rose from 20.7 % to 28.0 %. Removing 53 575 rendered points removed structure the projection found easy, so distance on screen means less than it did. Neither the regions nor the matching ever used those positions.

7.6 Where the descent starts: cluster median against an amortised inverse

Coordinate descent starts each type's gesture axes at their defaults, which are wrong for 78 % of types on `gammaScale` alone. Since we own the forward model, training data for an inverse is free: sample a parameter vector, render it, and measure the render with the same patch function the critic consumes. This is the encoder-decoder arrangement of differentiable DSP [17] with a physical decoder, which is why the interpretability survives.

The gate was written before the encoder was trained, and both predicted weak spots were recorded before the run: `tractQ` is the worst-learned head by an order of magnitude (validation MSE 1.125 against `gammaScale`'s 0.102) and timbre classification reaches 0.442 over three classes where chance is 0.333.

The median and the mean both rise, so the encoder does something real; it improves barely more types than a coin would, and it costs one species a tenth of its fit. It is not adopted, and the flag is off by default.

The finding that matters is diagnostic. *The starting point was not the binding constraint.* Coordinate descent from the cluster median already reaches a comparable basin about half the time, which is more than the plan assumed when it described the search as unable to escape a bad basin. So the remaining fit error is mostly not a basin problem, and what is left of "the model is right and the fit is what fails" lives in the objective and in the physics beyond the oscillator. The mechanism of the encoder's losses is visible in a separate arm: an encoder-only start scores *below* the cluster median it replaces (0.342 against 0.442), because on the fields the median already measures — timbre, attack, hold, harmonic, the modulations — a regression estimate is worse than a measurement. Its only new information is on the four gesture axes the median carries no value for, and the wins cluster exactly there.

A two-start variant was pre-registered in the same file rather than re-running the gate with a softer bar: run the descent from both starts and keep whichever scores better on the objective that already decides, which cannot regress by construction. Its bar is +0.02 median with *no* species worse at all. Its measured cost is roughly 61 minutes against the single-start arm's 11, because the encoder start explores a different basin and its descent does not terminate as early.

It passes, and it is the only pre-registered gate the inverse encoder has cleared. Median resynthesis cosine rises 0.7410 → 0.7740, a gain of +0.0330 against the +0.02 bar; 51.7 % of the 1 661 shared types improve, **none regress**, and no species median falls (87 up, 13 unmoved, 0 down). Debug-100 subset, scored through the fit's own per-type log against the baseline arm.

The route there is worth reporting, because the gate's second condition earned its keep. The first two-start run cleared the median bar at +0.0320 and *failed* on 48 of 1 661 types coming out worse, by up to 0.049. Since a two-start cannot regress by construction, the pre-registration had already named that outcome a bug in the selection rather than a trade, so it was diagnosed rather than excused. Three candidate explanations were eliminated by measurement — the search objective and the reported metric are the same computation; the fricative refinement is monotone in its own input; the two arms fitted identical inventories — and a fourth by direct test: the renderer is bit-identical for fixed parameters, so the reported cosine is a value and not a draw. The cause was that the selection compared candidates *before* the fricative refinement while the shipped and gated number comes *after* it, and that refinement's benefit depends on the basin: 19.3 % of types carry the fricative flag, and 100 % of the 48 re-

Table 3: The same four arms scored on one query population: the 160 279 field points every arm holds unchanged. Label assignment is worth +4.3 points and is the same in both anchor conditions, so the effect is the clustering's. Removing the synthetic anchor is worth nothing on this statistic; what it is worth is half the regions and a quarter fewer points. Debug-scale.

arm	points	k	field-point type retrieval	clustering (s)
k-means, as shipped	214 154	4 298	0.7176	54.8
label assignment	214 154	4 298	**0.7604**	0.2
label assignment + no synthetic anchor	160 579	2 155	**0.7606**	0.1
k-means + no synthetic anchor	160 579	2 155	0.7156	15.2

Table 4: The amortised inverse encoder as an initialiser, on two `learn` runs over the 100-species debug store differing only in one flag. 1 661 fitted types, all shared. The gate was stated in full before the encoder was trained and fails on all three conditions. Debug-scale.

	baseline	encoder	gate	
median resynthesis cosine	0.7411	0.7546 (+0.0135)	$\geq +0.02$	**fail**
mean resynthesis cosine	0.6824	0.6997 (+0.0173)	—	
share of types improved	—	877/1 661 = 52.8 %	$\geq 60\,\%$	**fail**
worst species' median change	—	−0.0816	≥ -0.05	**fail**

gressions did. Scoring each candidate as it would ship took the regressions to zero and the median gain from +0.0320 to +0.0330.

Two limits belong with the number. It licenses a fit option that is never worse; it does not license an audible claim, because the gate reads the same 28-d cosine whose correlation with spectral distance is about 0.1 (§7.3), so a bench run on cached audio is what an audible improvement would need. And the single-start arm's published worst-species figure of −0.0816 was scored through the same pre-refinement comparison, so part of it is that artefact rather than the encoder's fault — which does not rescue that arm, since it failed on the median alone.

7.7 Grammar order on held-out bouts

Four models, one protocol: per species, an 80/20 split by *bout*, a k-order backoff model fitted on the train half for $k = 0 \ldots 3$, every held-out token scored, cross-entropy in bits per token. The corpus is 34 696 bouts, 327 164 tokens and 2 421 species, of which 217 clear the 20-bout bar. The control is the same corpus with each species' tokens redealt within species, keeping bout lengths and marginals. The loader is pinned against the shipped bout function and refuses to run if the bout rule has drifted.

This table is the source of both the strongest positive result in the paper and one of its clearest negatives, and Section 9 reads it in full.

A separate bake-off compared model *classes* at first order and above. A repeat-count-dependent model in the style of Jin and Kozhevnikov [34] lost decisively at −0.348 nats per token, winning 0 of 30 species. A hidden-state HMM won the pre-registered rule at +0.422 nats and 29 of 30 species — and the decomposition reversed the adoption. Splitting held-out tokens by whether the transition was seen in training, first-order is the better model on the 90 % seen (−1.525 against −1.624 nats/token, better in 15 of 30 species), and the HMM's entire advantage is the 10 % unseen tail, where first-order pays −9.25 and the HMM −4.16. That is a property of smoothing *shape* — a mixture predictive is heavy-tailed — not of recovered hidden structure. Adopting a heavier model for a smoothing artefact would be the wrong lesson, so first order stayed and the smoothing was repaired instead: a Kneser–Ney continuation backoff [37] gains +1.90 and +2.32 nats per token on unseen-*context* transitions across two debug stores with the seen side bit-identical, and passes its own separately pre-registered gate. Two honest caveats: 80 % of the unseen tail is seen-context with an unseen continuation, outside the repair's scope, so the full-tail gain is +0.37 and +0.65; and the repair narrows the HMM's advantage, which confirms the diagnosis.

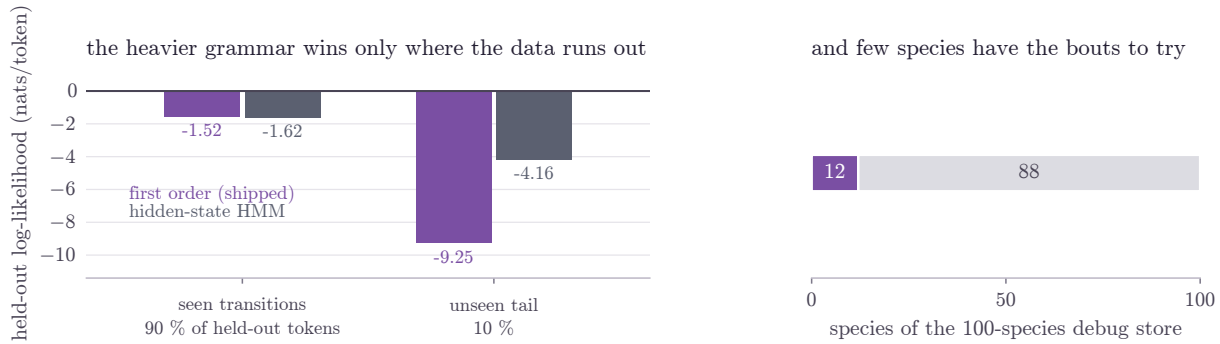
After the depth harvest the bake-off was re-run with qualifying species at 104 rather than 30. The HMM won again, at +0.4976 nats and 78 of 104 species, and this time against a partially repaired baseline: subtracting the continuation backoff's own measured gain on the same splits leaves the HMM ahead by about +0.37 nats per token. The debug-era explanation cannot account for a residual that size, so something closer to genuine hidden structure is scoring better now that the data can support it. It has *not* been adopted: the shipped inventory grammar is raw first-order counts, there is no HMM-state grammar writer, and adopting one changes what the fit emits, what the application improvises from and what song typing aligns against. That is recorded as the next syntax task with the number to beat.

7.8 Song-level structure: an edit-distance kernel against bag-of-types

Whole bouts as points, as (token, gap) sequences under a weighted edit-distance kernel with the rhythm term priced inside the dynamic program, judged against plain bag-of-types over the same bouts. It fails

Table 5: Held-out cross-entropy by Markov order, real corpus against its within-species shuffle. 34 696 bouts, 327 164 tokens, 2 421 species; 217 species clear the 20-bout bar. The gate — median gain ≥ 0.05 bits/token at order $1 \to 2$, ≥ 0.05 more than its own shuffle, and share of species gaining $\geq$ control +20 points — fails on the magnitude criterion and passes on the share criterion, at three seeds.

	real	control
held-out cross-entropy, $k = 0$ (unigram)	3.284	2.879
$k = 1$	**2.301**	3.634
$k = 2$	2.546	4.868
$k = 3$	2.866	5.313
median gain, order $1 \to 2$ (bits/token)	**−0.091**	−0.912
median gain, order $2 \to 3$	−0.108	−0.187
share of species gaining at $1 \to 2$	**25.3 %**	4.1 %



Left: first order is the better model on the 90 % of held-out tokens whose transition was seen in training, and the hidden-state model's entire advantage is the unseen tail. That is a property of smoothing shape, not of recovered hidden structure, so the smoothing was repaired instead: a Kneser-Ney continuation backoff gains 1.90 and 2.32 nats per token on unseen-context transitions across two stores. Right: twelve species reach 20 bouts after run-collapse, which is the syntax threshold. Counted before the depth harvest; the production re-count is 217 of 2 421 species. Both panels are debug-scale.

Figure 13: Left: held-out likelihood split by whether the transition was seen in training. The heavier grammar's whole advantage lives where the data runs out. Right: species of the 100-species debug store with enough bouts to attempt song-level structure, counted before the depth harvest. Both panels are debug-scale.

both pre-registered gates on both debug stores: species nearest-neighbour lift +0.877 against bag-of-types' +0.924 on the 37-species store and +0.050 against +0.081 on the 100-species store, with recordist lift +0.264 and +0.087 against a < 0.05 bar. Three measured details survive the failure. Within-species song structure exists — all 18 deep species separate their own recordings — and lives entirely in the token marginals, so order and rhythm add nothing measurable at current depth. The rhythm term actively *costs* species lift. And the bar for any learned song encoder is bag-of-types, not the edit kernel.

7.9 Model-free sequence statistics against their own shuffle

Three statistics that fit nothing, run on the production corpus against the same within-species permutation control. Conditional entropy of the next token given one previous is 0.941 bits against the shuffle's 3.343 — not a subtle effect. The Menzerath–Altmann exponent's median is +0.019 where the law predicts negative, with 45.7 % of species negative against the control's 8.5 %: a weak effect in the law's direction, not the law, and not the shape reported for geladas or penguins [19, 28]. The control's own exponent is consistently positive, which is a sampling artefact rather than a finding — a mean over more draws sits closer to the species mean and its logarithm rises with the count — and that artefact is exactly why the control is run. Normalised compression distance [12] is refuted outright, because the control separates species *better* than the real corpus does (0.038 against 0.018) — compression distance here reads alphabet and marginals, which the shuffle preserves, and not order. No learned song encoder should be measured against it.

One arm from this set was retracted by the project itself and is recorded as a design error rather than as a result. Comparing the $k = 1 \to k = 2$ entropy drop between real and control is not a valid test when the two curves start 2.4 bits apart, and past $k = 3$ the control's plug-in curve falls *below* the real one because a plug-in estimator reports near-unique contexts as determined. Both effects are bias. The verdict function now refuses that comparison rather than returning the answer it would have returned, and the honest instrument — held-out cross-entropy, unbiased by construction — is Table 5.

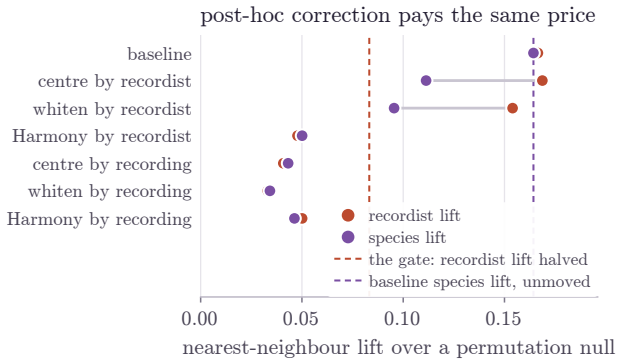
7.10 Detector configurations against a permutation control

Three configurations of the song detector were run against the same corpus-wide shuffle control, whose

yield bar was 0.2 % of candidate bouts.

Repeat classes pass on both halves of their gate and are adopted. The control is unchanged to the bout, because shuffling a species' tokens destroys runs, so the control finds no repeat classes at all — the sharpest evidence available that the ones found are real.

The two refuted arms are instructive in the same way. Letting each species set its own recurrence bar on the evidence of its own shuffled tokens let 2 398 of 2 421 species take the loosest bar, because a species with eight candidate bouts has a control that cannot place a bout at any size: "the control found nothing" meant "the control had nothing to work with". Adding a power requirement repairs the mechanism but not the verdict — no candidate threshold in $\{10, 15, 20, 30, 50\}$ returns the control to 0.2 %, and the rows that come closest gain almost nothing over the 98 species repeat classes already reach. A per-species bout rule at five times the species' own median gap reaches 105 species at nearly four times the chance yield, because splitting a fast singer's bouts finer produces more short strings and short strings are what chance can chain. Both were removed with their flags: a refuted policy nobody can reach is surface, not machinery, and the tables are what survive of them.



The gate asked for the recordist lift halved with the species lift not falling at all: the clay dot left of the clay rule and the violet dot still on the violet one. No arm does both. Subtracting each recordist's mean leaves the recordist lift higher than it started while costing a third of the species structure, so the confound is not a location shift.

Figure 14: Post-hoc batch correction of the production embedding reaches the confound and pays exactly the price a gradient-reversed adversarial head paid. The gate asked for the recordist lift halved with species lift not falling at all; no arm lands in it. Two unrelated mechanisms landing on the same exchange rate is evidence about the data rather than about either mechanism, and the fact that subtracting each recordist's mean leaves the recordist lift *higher* than it started says the confound is not a location shift.

7.11 What this section licenses

Read together, the internal comparisons say four things. The choice of *space* matters most and is settled: the learned embedding beats the hand-weighted vector by 6 to 13 points of species placement, replicated. The choice of *clustering* matters exactly as much as the region budget binds, and no more. The choice of *objective* is unresolved and is the project's central open problem: the best available ruler is the worst available optimiser. And the choice of *starting point* does not matter, which is a genuinely useful negative because it eliminates one of the two remaining explanations for the fit's failure.

8 Our methods against other approaches

8.1 What is and is not a fair comparison

The external comparisons in this section are all read-only tests on a fixed store, through one estimator — nearest-neighbour lift over a label permutation null, blocked by recording — with no training and no new data. That makes them directly comparable to every lift this project has published. Three limits apply and are stated before the numbers, because they change what the numbers mean.

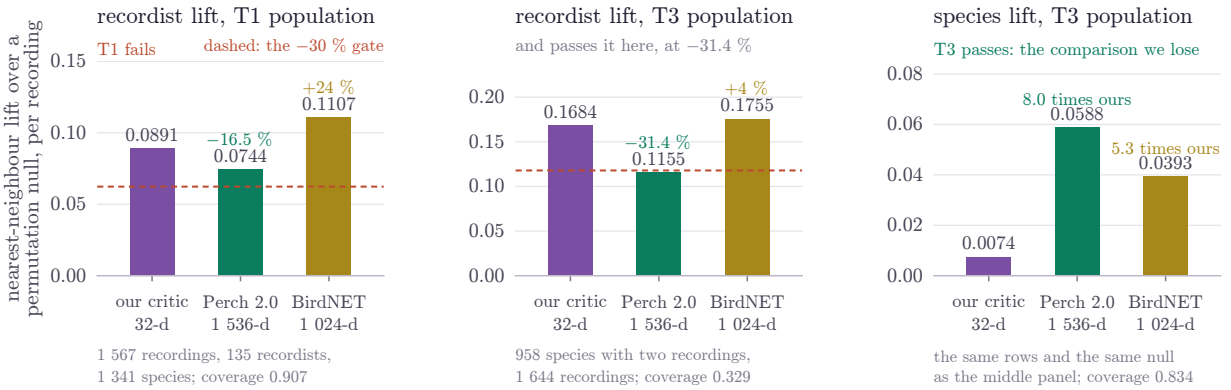
First, **the representation is per recording, not per syllable**. Perch's point is the mean of a recording's first six 5-second window embeddings; BirdNET's is the analogous average of its 3-second windows; the critic's is the mean of that recording's syllable embeddings. That form was chosen because two points from the same recording cannot exist in it, so same-recording leakage is impossible by construction rather than by exclusion. But the critic's embedding is a *syllable* space being averaged over a recording, which is not the job it was fitted for. So a species-retrieval result here says that an external space is the better *per-recording* descriptor. **It does not say that Perch or BirdNET beats our critic syllable for syllable. That comparison has never been run.** The reason it has not is recorded: a per-syllable Perch pass over 340 079 syllables is about 3.3 hours on this machine's CPU, the installed ONNX runtime does not match the installed CUDA version, and the GPU provider fails to load and falls back to CPU silently at no measured speed difference.

Second, **the populations differ between the two draws and the difference is the result**, not noise to be averaged away. One draw was built for the recordist question, one for the species question, and they disagree about the recordist verdict. The tie-break was fixed on coverage before the disagreement appeared.

Third, **the audio cache constrains what can be asked at all**. It holds at most two recordings per species — 2 616 species with exactly two, 1 221 with one, none with three — because the iNaturalist ingest took no more and the xeno-canto audio was deleted at harvest by design. No species group of eight recordings exists on any audio this machine holds, so a species-stratified draw returns zero recordings and no selection rule can produce one. Anything needing several recordings of one species needs audio re-fetched first.

Table 6: Song-detector arms on the production store (2 421 species, 34 696 bouts, 10 550 candidate bouts of three tokens or more). Only the first passes. The control yield is the share of candidate bouts a within-species shuffle places, and the pre-registered bar is 0.2 %.

arm	species with a song	song types	placed bouts	control yield	verdict
shipped, motifs only	48	96	1 052	0.170 %	baseline
+ **repeat classes**	**98**	**356**	**3 741**	**0.170 %**	adopted
per-species recurrence bar	99–202	452–636	—	0.266–0.442 %	refuted
per-species bout rule	105	449	—	0.628 %	refuted



Read-only tests on a fixed store, one estimator, no training and no new data. The representation is per recording in every row, so same-recording leakage is impossible by construction - which also means a species result here says an external space is the better per-recording descriptor, and does not say Perch beats our critic syllable for syllable, a comparison that has never been run. T1's verdict rests on a coverage tie-break: it passes at −31.4 % on the T3 population and fails at −16.5 % on its own, whose recordist estimate covers 0.907 of rows against 0.329, so the instrument built for the question wins. The finding that holds in both draws is the one to carry: the recordist lift is twice the species lift in Perch's space, 4.5 times in BirdNET's and 23 times in ours. Every space tested is confounded in the same direction and they differ only in degree.

Figure 15: Three spaces, two populations. Recordist and species nearest-neighbour lift over a permutation null, per recording, for our 32-dimensional critic and two archive-scale external models. Perch 2.0 — trained on the same archives at roughly a hundred times our scale and never asked to tell our resynthesis from a bird — still carries five sixths of the recordist structure ours does, and BirdNET carries more than ours in both populations. If archive-scale training conferred recorder invariance the two largest models trained on these archives would both have it and they would agree; one is better than ours and one is worse, which locates the property in the archives. The right panel is the comparison we lose, and lose clearly. Hue carries model identity here, not a verdict.

8.2 Recording-condition invariance: three spaces, two populations

T1 fails, and it is the fourth independent closure of the same road. Perch 2.0 — trained on the same archives at roughly a hundred times our scale, never asked to tell our resynthesis from a bird, and therefore structurally incapable of having learned the axis that splits our types in two — still carries five sixths of the recordist structure our own critic carries. The gate asked for −30 % and measured −16.5 %. The prediction was written in advance and is worth quoting rather than softening: *if a model trained on the same archives at a hundred times the scale is no more recorder-invariant than ours, the confound lives in the archives rather than in our critic.* It does.

BirdNET fails harder, and the disagreement between the two external models is stronger evidence than either alone. BirdNET's recordist lift is *above* our critic's in both populations (+24 % and +4 %). Perch reads −16.5 % and BirdNET +24 % on the same rows. If archive-scale training conferred recorder invariance, the two largest models trained on these archives would both have it and they would agree. One is better than ours and one is worse, which locates the property in the archives rather than in any model's loss — the same conclusion the batch-effect analysis reached from the inside, now reached from outside with two independent instruments.

T3 passes decisively, within its stated scope. Perch retrieves a species' other recording 8 times better than our critic does (0.0588 against 0.0074) and BirdNET 5.3 times better (0.0393), on the same rows against the same null. The direction replicates across both external models, so "an external space is the better per-recording species descriptor" holds. It licenses Perch as the per-recording batch descriptor for any future invariance work, and it weakens the case for building further external ports to answer T1, which is answered three ways.

The finding that holds in both draws is the one to carry. In Perch's space the recordist lift is still twice the species lift; in BirdNET's it is 4.5 times; in ours it is 23 times. Every space tested is confounded in the same direction and they differ only in degree. A model trained on these archives at a hundred times our scale, which has never seen a resynthesis, still encodes *who recorded it* more strongly than *which bird it is*.

Two things are explicitly not settled. T1's own verdict rests on a coverage tie-break: it passes at −31.4 % on the T3 population and fails at −16.5 % on the T1 population, and the T1 population's recordist estimate

Table 7: Recordist and species nearest-neighbour lift over a permutation null, for our 32-dimensional critic and two archive-scale external models, on two populations. Per recording in every row; same rows and same null within each population. *Coverage* is the share of rows for which the estimate is defined. T1's gate asked for a 30 % relative reduction in recordist lift; T3's asked for a species-retrieval improvement.

space	dims	recordist lift	species lift
T1 population: 1 567 recordings, 135 recordists, 1 341 species; recordist coverage 0.907			
critic	32	0.0891	not estimable
Perch 2.0	1 536	**0.0744** (−16.5 %)	not estimable
BirdNET	1 024	0.1107 (+24 %)	not estimable
T3 population: 958 species with two recordings each, 1 644 recordings, 760 recordists; species coverage 0.834, recordist coverage 0.329			
critic	32	0.1684	0.0074
Perch 2.0	1 536	**0.1155**	**0.0588**
BirdNET	1 024	0.1755	0.0393

covers 0.907 of rows against the other's 0.329, so the instrument built for the question wins and the verdict is fail. And the novelty comparison — Perch against our critic's novelty ROC of 0.846 — was never run. Retiring the critic's species head was made conditional on T1 *and* T3 landing; T3 landed and T1 did not, so it stays.

8.3 Post-hoc batch correction against training-time invariance

The same confound was attacked with the genomics batch-effect toolkit, which has the property that its strongest members are post-hoc: they correct an embedding that already exists. 20 000 syllables of the production embedding, 885 species, 1 001 recordists, 3 265 recordings, every arm scored by the same estimator against its own null.

Three findings, in increasing order of consequence. The confound is not a location shift: subtracting each recordist's mean leaves the recordist lift higher than it started while costing a third of the species structure, which is a mechanistic explanation for why two training-time attempts failed and one that was never available before, because both of those attacked the confound during training where nothing can be inspected. Harmony [38] reaches it and pays exactly the price a gradient-reversed adversarial head paid: −71 % recordist, −70 % species. Two unrelated mechanisms landing on the same exchange rate is evidence about the data rather than about either mechanism. And correcting by recording removes both because the labels are nearly nested — a recording normally holds one bird of one species — which is a statement about how the corpus was collected.

Against that, the data road works and its effect size shrank honestly when re-tested. Figure 16 shows both scales. Merging two genuinely different recording domains — iNaturalist soundscape and xeno-canto focal — bought +14.4 points of cross-domain species retrieval on the 12 species present in both stores, with a 3-species holdout. The pre-registered bigger re-test on 63 both-domain species with 16 held out took that to +3.6. A cross-domain contrastive mechanism on top reached +13.4 at the small scale and +1.7 against a +10 gate at the larger one, so the mechanism's boost was small-sample structure. The uncomfortable part is recorded plainly: the merge's *own* win shrank too, and its held-out realism AUC reads below the single-domain arm's here (0.976 against 0.995) where the first test gained 0.012. Every number from the first test should be treated as small-sample in both directions. What survives the bigger test is the ordering, which is the load-bearing claim: more domains per species helps, and the mechanism adds approximately nothing on top of the data.

8.4 Positioning

Table 9 places the system beside the approaches it is most likely to be confused with. A cell reads "not documented" where the cited sources do not answer it; the column entries for this work are properties of the shipped artefact, not aspirations.

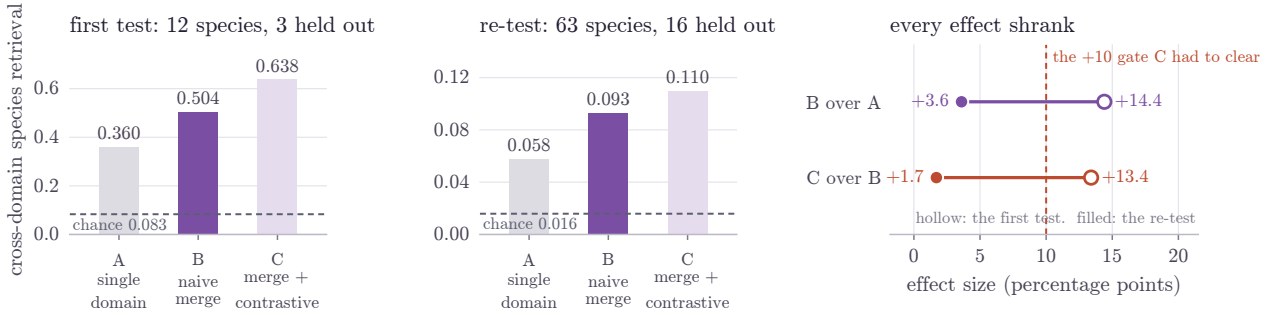
The comparison that matters is not on accuracy, because the systems answer different questions. The external models are better at "which bird is this", by a margin that has been measured here at 5 to 8 times on per-recording species retrieval, and this work should be read as taking that for granted rather than competing with it. What this work provides that none of them does is a *generative, physically parameterised* account of a syllable that is fitted to field audio at archive scale, together with an explicit statement of the evidence behind every claim built on it.

9 What the data supports

This section states what can be read out of the corpus, at what strength, with the ceiling attached to each claim. Numbers from the 100-species debug subset are marked *debug-scale*; numbers from twelve-species or ten-type samples are marked as such, because the project has twice measured an effect at small scale that did not survive replication, in both directions.

Table 8: Post-hoc batch correction of the production embedding. 20 000 syllables, 885 species, 1 001 recordists, 3 265 recordings. The gate asked for the recordist lift halved with species lift not falling at all. No arm passes: every arm that halves one halves the other.

arm	species lift	recordist lift	ratio
baseline	0.1643	0.1665	0.99
centre by recordist	0.1113	0.1687	0.66
whiten by recordist	0.0955	0.1540	0.62
Harmony by recordist	0.0501	0.0478	1.05
centre by recording	0.0432	0.0408	1.06
whiten by recording	0.0342	0.0331	1.03
Harmony by recording	0.0463	0.0501	0.92



Merging two genuinely different recording domains - iNaturalist soundscape and xeno-canto focal - is the data road; the contrastive mechanism on top is the method road. Hollow marker in the right panel: the first test. Filled: the pre-registered re-test. The mechanism's boost was small-sample structure and the data road's was not, though it survives at a quarter of the size. The merge's own held-out realism AUC reads below the single-domain arm's in the re-test (0.976 against 0.995) where the first test gained 0.012. Both panels are debug-scale and absolute levels are comparable only within a panel.

Figure 16: Cross-domain species retrieval for three training arms at two scales. Both panels are debug-scale and absolute levels are comparable only arm against arm inside each panel. The pre-registered re-test shrank every effect; the ordering survived.

9.1 The tokens are units. This is the strongest result in the paper.

A first-order transition model over the learned syllable-type inventory buys **0.98 bits per token** on held-out bouts — cross-entropy 3.284 down to 2.301 — while on a within-species shuffle of the same tokens, with bout lengths and marginals preserved, the same model *costs* 0.75 bits (2.879 up to 3.634). The corpus is 34 696 bouts, 327 164 tokens and 2 421 species; the estimator is unbiased by construction, which the plug-in estimator this replaced was not.

The logic is what makes it strong. An unbiased estimator paying for parameters it cannot use is the clean form of the claim that the tokens are units: the model can only win if the transitions are real, and it wins by a bit per token. This holds a load-bearing assumption that everything above the syllable rests on. The syllable inventory is produced by k-means in a learned embedding, which is a clustering; the question of whether it is also a *unit system* is not answerable by comparing two models, and it is answered here in the affirmative. A model-free estimate on the same corpus agrees: conditional entropy of the next token given one previous is 0.941 bits against the shuffle's 3.343.

Two artefacts belong with the number. The control's unigram cross-entropy sits *below* the real corpus's (2.879 against 3.284), because shuffling spreads each species' tokens evenly across its bouts so that train and test marginals nearly coincide, whereas real bouts are clumped — a bout tends to repeat one type — so a held-out real bout can be mostly a type the train half barely saw. That clumping is itself structure, and it is why the gate reads within-corpus deltas rather than the two absolute curves against each other. And the same behaviour is pinned in a self-test: a planted second-order process must be found, a first-order process must not show depth, and the shuffle of the planted process must not gain from a longer context.

9.2 Repeat classes double the corpus's songs, with the control unmoved

The song funnel was counted stage by stage on the production store, and the loss is accounting rather than machinery. Of 340 079 syllable observations, 327 164 (96.2 %) are typed. Those form 34 696 bouts. **19 738 of those bouts (56.9 %) collapse to a single token**, and 6 940 of them are three syllables or more — a bird repeating one sound, which is a repertoire fact and not a measurement error. Only 10 550 bouts (30.4 %) survive as candidates, 366 species have enough of them to ask the question, and 48 species got a song type.

The collapse is correct for a motif distance — a trill sung twice at two speeds must not sit twenty edits apart — and wrong as a filter. Until this was fixed, *a bird whose song is a trill could not have a song type at any recurrence bar.* Adding repeat classes takes the corpus from **48 species with a measured song to**

Table 9: Approaches beside each other. External rows are read from the cited sources; "not documented" means the source does not answer the cell. *Interpretable parameters* means a reader can name the quantity that produced a given sound.

approach	what it is	sings	interpretable parameters	runs as a static asset set	labels its claims
BirdNET [35]	species classifier on spectrograms	no	no	no	not documented
Perch 2.0 [73]	general bioacoustic embedding, ∼15k classes	no	no	no	no
AVES [29]	self-supervised animal audio encoder	no	no	no	no
BirdMAE [59]	masked autoencoder for bird audio	no	no	no	no
NatureLM-audio [64]	audio-language model for bioacoustics	no	no	no	no
Sainburg et al. [66]	projected latent space across repertoires	no	latent, not physiological	not documented	no
Goffinet et al. [25]	learned low-dimensional vocal features	no	no	not documented	no
generative audio models	waveform generators	yes	no	no	no
Lyrebird	physical syrinx + a corpus with no audio in it	**yes**	**yes** — pressure, tension, tract geometry	**yes** — no audio retained, no weights shipped	**yes** — five levels

98, song types from 96 to 356 (96 motifs, 260 repeats), and placed bouts from 1 052 to 3 741 — with the shuffle control **unchanged to the bout at 0.170 %** against a 0.2 % bar.

The control is the reason to believe it. Shuffling a species' tokens destroys runs, so the control finds no repeat classes *at all*. A statistic whose null yields zero, on a corpus where the real version yields 260 types, is about as sharp as evidence gets from a permutation design. And the comparison is a comparison: the baseline arm reproduces the shipped corpus exactly, at 48 species, 96 types and 1 052 placed bouts.

The 0.2 % bar those arms are judged against was itself set on a control sweep rather than chosen, and the sweep is worth quoting because it shows the shape of the trade-off. On an earlier 35-species store, lowering the minimum cluster size from 6 to 4 took the real yield from 50 song types to 113 and the control's from 1 spurious type (6 of 3 386 bouts, 0.2 %) to 17 (103 of 3 386, 3.0 %). Raising it to 8 gained the control nothing and lost half the real types. Six is the knee, and the residual 0.2 % is reported rather than tuned to zero, because the price of zero was half the signal. Two further constants exist for measured reasons: bouts collapsing below three tokens are set aside because chance two-token strings sit at normalised edit distance 0.5, and a cluster whose members' weighted mean distance to their medoid exceeds 0.5 is demoted to noise, because five random six-token strings were once reported as one song type.

The ceiling is equally clear. Ninety-eight of 2 421 species is 4.0 %. Two attempts to raise it by loosening a threshold both bought chance strings faster than songs (Table 6). The remaining route is depth, and depth has an archive ceiling.

9.3 A second order is not supported, and 217 species can be asked

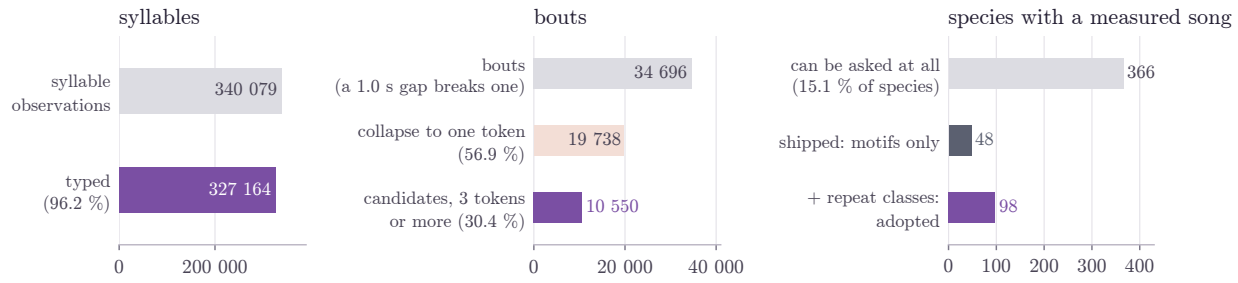
The median species *loses* 0.091 bits per token going from order 1 to order 2 on bouts it has not seen, at three independent splits (−0.091/ − 0.116/ − 0.142). For the median species a second-order model is worse out of sample: the corpus cannot estimate the contexts it asks for.

The share criterion tells the other half. 25.3 % of species gain something at order 1 → 2, against the shuffle's 4.1 % — a six-fold rate that clears the pre-registered 20-point margin. So the honest single sentence is: *there is no second-order structure this corpus can support, and a quarter of species hint at one.* The hint is not nothing; it is also not what the gate asked for, and the gate was written first.

What this settles is the diagnosis. The depth question is **data-limited, not method-limited**: 217 of 2 421 species have enough bouts to ask it at all. Given that the literature is unambiguous that birdsong is not first-order [34, 44], the 25 % that hint at a second order are an argument for depth on more species rather than for a cleverer sequence model. The post-depth-harvest bake-off supports the same reading from the other side: with qualifying species at 104 rather than 30, a hidden-state model beats even a smoothing-repaired first-order baseline by about +0.37 nats per token, where at debug depth its entire advantage was smoothing shape.

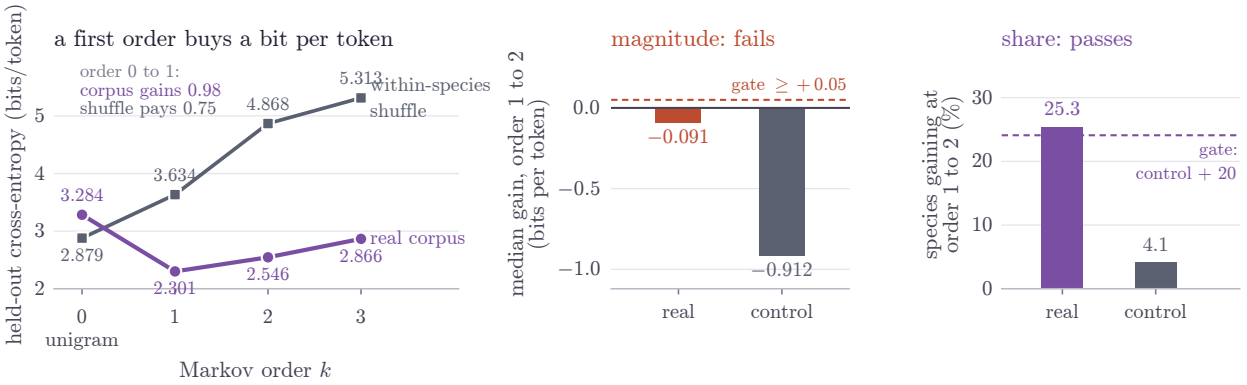
9.4 The map's own confusion is the honest bound on the interaction

With every type granted its own cluster, so that no type can fail to be retrieved for want of one, **type retrieval is 0.602**. For 40 % of points the nearest



The funnel is drawn in the three units it actually changes through rather than on one axis. The loss is accounting, not machinery: 6 940 of the collapsing bouts are three syllables or more, a bird repeating one sound, which is a repertoire fact and not a measurement error. Repeat classes - one collapsed string heard at least six times, matched by equality - take song types from 96 to 356 (96 motifs, 260 repeats) and placed bouts from 1 052 to 3 741, with the within-species shuffle control unchanged to the bout at 0.170 % against a 0.2 % bar. Ninety-eight of 2 421 species is 4.0 %. Production store.

Figure 17: The song funnel on the production store, drawn in the three units it actually changes through rather than on one axis. Of 340 079 syllable observations, 327 164 are typed and form 34 696 bouts, of which 56.9 % collapse to a single token and 30.4 % survive as candidates; 366 species have enough candidates to ask the question and 48 got a motif song type. Adding a repeat class — one collapsed string heard at least six times, matched by equality — takes that to 98 species with the permutation control unmoved to the bout.



Lower is better on the left. Per species, an 80/20 split by bout, a k-order backoff model fitted on the train half, every held-out token scored. An unbiased estimator paying for parameters it cannot use is the clean form of the claim that the tokens are units: the model can only win if the transitions are real. A second order is not supported - the median species loses 0.091 bits per token at three independent splits - while a quarter of species gain something, six times the shuffle's rate. 217 of 2 421 species clear the 20-bout bar at which the question can be asked. Production scale.

Figure 18: Held-out order scaling. A first-order transition model over the learned syllable-type inventory buys 0.98 bits per token on bouts it has not seen, while on a within-species shuffle of the same tokens, with bout lengths and marginals preserved, the same model *costs* 0.75 bits. The estimator is unbiased by construction, so a model can only win if the transitions are real. A second order is not supported: the median species loses 0.091 bits per token, which fails the magnitude criterion, while a quarter of species gain something, which passes the share criterion. Production scale.

centroid by cosine — the exact lookup the browser performs on live audio — belongs to *another type*. This is debug-scale, on a fixed 214 154-point matrix over 2 143 types.

This is the number to quote when asked whether singing at the system lights the right region. It is not the clustering's failure and it is not the projection's: it is the embedding's own confusion between types, and it was invisible under k-means because a type with no cluster cannot be retrieved at all. Raising it is a representation problem, which is where Section 8's external comparison points.

Two related numbers complete the picture and are better than the headline suggests. On the field points alone — the population the question is about, since a recorded syllable must light its own region — retrieval is 0.7604 under label assignment and 0.7176 under the shipped k-means (Table 3). And the browser's bridge, which is a second and independent source of error, loses 38.6 % of region agreement of which 76.4 % is boundary near-ties: tie-adjusted agreement is 0.91, and species-level agreement, which is the honest single-label claim, is 0.781 against chance 0.033.

9.5 Pathological recording sessions are detectable from the browser's side

Eight of 45 held-out recordings carry 22.9 % of the map's total flip loss, all eight from xeno-canto and none from iNaturalist, with one session alone carrying 9.3 %. They separate cleanly on a statistic the browser can compute from the 28-dimensional side alone: the mean 5-nearest-neighbour distance of a session's standardised descriptors to the fit-side cloud separates them at **AUC 0.953**, and a shipped cloud of 256 random anchors holds AUC ≥ 0.92 over draws at a cost of 28 kB. The pre-registered adoption gate for using it as a *lever* — raising the remaining rows' agreement by 5 points —

failed at +3.5 points raw, so the pre-registered fallback fired and it ships as an *honesty* gate: the interface says the session is off everything the map knows instead of lighting regions. At the shipped operating point it flags 8 of 45 sessions, catches 5 of the 8 bad ones including the three worst loss-carriers, and produces 3 clean false flags all within 0.02 of the bad line.

Two calibration facts belong with it. A threshold calibrated on fit sessions does not transfer to unseen sessions even when the session's own rows are excluded, because sibling sessions of the same recordists deflate every fit session's distance — the 95th-percentile fit-side threshold flags 29 % of held-out sessions. The shipped threshold is calibrated cross-wise, each fit session judged against the other seeded half. And the mirror-image action fails: pruning the sessions this signal flags from the *fit* costs −0.0147 and −0.0062 of evaluation cosine on two stores against a 0.005 allowance. The pathological sessions are manifold coverage, and removing them hurts the next unseen pathological session.

9.6 The physical model is right and the fit is what fails

The audit's conclusion holds and is unusually well supported for a negative: *nothing the project has measured points at the oscillator.* The ODE is the published normal form term for term, agreeing with Runge–Kutta to 2 % of f_0, reproducing the reference measurements to 0.1 % on f_0 and 2 % on amplitude, holding at 18 of 18 two-sided parity fixtures, and collapsing into period doubling exactly where the coupling literature says the weak-coupling approximation dies. Every recorded deviation is in parameterisation, in the periphery, or in the objective.

The concrete state of the fit, at production scale: median resynthesis cosine **0.648** over 2 421 species and 13 859 types, with the worst type at −0.445. The path to that number is worth stating because it is not monotone and the reasons differ: the 12 724-type learn scored 0.650, the per-species respiration and two-voice fit passes took it to 0.6623 across 12 680 scored types, and it then fell to 0.648 when the corpus deepened, which is the expected direction — the added species are harder. On the original 22-species store the equivalent figure reached 0.775 after the naturalness fields were added, from 0.533 before, with the share of types above 0.7 going from 26.2 % to 66 % and the share below 0.5 from 44.2 % to 11 %.

The single most stubborn number in the project is cross-species placement. On the original 22-species store a resynthesis lands nearest its *own* species 23.1 % of the time against chance 4.5 %. Four hypotheses for the remaining 77 % were each tested and each refuted, and they are worth listing because each cost real work: the envelopes are not collapsed (we express 103 % of the real between-species spread and 106 % of the spectral detail — they vary along different axes); there is a systematic spectral tilt and correcting it *exactly* drops placement to 20.2 %; the harmonic structure is not missing, because real syllables resolve to a mean of 2.2 partials and feeding measured partial amplitudes to the fit moved the cosine 0.776 to 0.768; and a time-resolved objective improves its own metric by 36 % while dropping 28-dimensional placement from 61.7 % to 43.2 %.

That last refutation is the informative one, and its mechanism was found afterwards. The two objectives conflicted because the f_0-tracking tract locks spectral shape to pitch, so the only lever the fit had for moving a spectral trajectory was bending the pitch contour — and the pitch contour is strongly species-diagnostic. Under a fixed waveguide the conflict disappears and all three metrics improve together. That the waveguide then failed to generalise across 22 species, while winning under learned typing on a different store, is recorded as unresolved.

One further reading of the placement metric is required, and it is the reason it is not the paper's headline. *Placement degrades in absolute terms as the corpus broadens, while improving relative to chance.* On the ingest that took the store from 36 to 121 species, own-species placement fell from 24.6 % to 11.5 % — while chance fell from 2.8 % to 0.9 %, so the ratio to chance rose from 8.8-fold to 12.8-fold. The pre-registered target for that gate was an absolute 35 %, and it **failed**. Own-type placement held at 62.6 % and the median cosine held, barely, at 0.742 against a 0.74 bar, with the reason named at the time: 85 of the newly added species carried about three recordings each, which fits thin types. Type coverage in the same build fell from 94.6 % to 54.1 % and cluster purity from 56.7 % to 30.3 %, and both are harvest depth rather than machinery: 651 orphan types belong mostly to those three-recording species, which cannot meet the map's minimum cluster size. Any absolute placement or purity figure in this paper is therefore only interpretable against its own species count.

9.7 Extensions ship only where a fit wins them, and the ones that ship are countable

The pattern is consistent enough to state as a result about method. **Hand-set physical defaults lose bake-offs, however correct the published physics.** The respiratory oscillator's hand-set default set moved median cosine from 0.343 to 0.336, with losses concentrated in sub-50 ms notes whose onsets the physically correct 20–40 ms attack eats. The two-voice coupling's hand-set default moved it from 0.343 to 0.294 — while the two worst-fitting low-pitched species *improved*, which is why it was kept rather than deleted. Both are parked off by default.

The per-species fit passes are the version that wins, and they win narrowly and specifically: respiration took 445 of 12 680 scored types and the second voice 2 027 of 5 922 screened types, with the ship gate holding everywhere. The tracheal comb won 10 of 22 species on the original corpus. The fricative route — turbulence routed around the f_0-tracking band into the syllable's own fitted formants, because a fricative is a noise burst

filtered by the tract and not a tone with noise added — moved the bench from 0.584 to 0.656 with 13 wins and 0 losses, and ships per type. A one-tone re-fit of the authored calls against the same bird's learned types shipped a measured noise/roughness texture to 14 of 15, median gain +0.148, with the honest caveat that the median cosine afterwards is still 0.270.

Every one of those gates was read on the descriptor cosine, and Section 6 says that cosine does not rank audible fidelity. A re-judgment of the shipped set on the spectral bench is therefore owed, and is recorded as owed.

9.8 Novelty detection improves with corpus breadth, and its threshold does not transfer

Novelty detection is a distance question in the embedding, and it is evaluated by holding entire species out of training and asking whether their syllables are flagged. The honest figure improved monotonically as the corpus broadened, and every step changed the test as well as the model, so the four numbers are a trend rather than a paired comparison. On the 9 000-syllable store with six species held out: ROC AUC 0.596 at 7.4 % true positives at the 5 % false-positive operating point. On the 47 000-syllable retrain with five held out: 0.672 and 14.8 %. On the 121-species store, evaluated against 25 892 syllables of 105 genuinely non-training species rather than a handful of chosen holdouts: 0.786 and 22.3 %. On the production critic, with 895 whole species held out — the hardest test run: **0.846 and 32.2 %**. The novel side of the test got larger and more honest at each step, and the doc that reports each pair says so.

An earlier and much higher figure, ROC 0.825 on the 9 000-syllable store, is a different measurement — held-back syllables of training species against held-out species' syllables — and was superseded in place by the 0.596 above. Both are on the record. The low-false-positive end of the curve is the weak end in every version.

The per-species pattern refutes the obvious explanation. Detection rate is not about taxonomy. On the 9 000-syllable store it fits five of six held-out species — congeners are harder — and the sixth contradicts it, because a wren with no relative in the training set is the *worst* detected of the six at 8.2 %. The ordering differs on the 47 000-syllable store, which is itself the point: the embedding measures acoustic neighbourhood, not relatedness. A loud broadband trill sits near other species' trills whatever the family tree. That is arguably the correct behaviour for the question being asked, and it means "hold out a species" is a harder test than "hear a genuinely new sound".

The operational finding is a warning. *A novelty threshold holds only for the corpus that calibrated it.* A threshold calibrated for 5 % false positives on the 9 000-syllable store flagged 29.8 % of the 47 000-syllable store, including 29 % of *training* species' syllables. Recalibration without retraining returns the store-wide rate to 5.6 %. The shipped threshold has moved through 0.1048, 0.1558, 0.1912, 0.1737 and 0.1323 across five harvest epochs. Thresholds do not transfer, and treating one as a constant is how a map starts confidently reporting novelty about sounds it already knows.

9.9 One example of ground truth from outside, at n = 31

Vocal mimics are the only source of labelled cross-species acoustic similarity available to a corpus like this one, because a lyrebird singing a grey shrike-thrush produces syllables whose acoustics are shrike-thrush and whose label is lyrebird — and the accuracy of the imitation has been certified by the receiving birds rather than by us [14].

The pilot is a well-formed test on an empty corpus, and is reported as such. The store holds **31** lyrebird syllables. Against the 12 393 type centroids of the production fit, those 31 match **13 distinct non-lyrebird species**, where random syllables of the corpus match 29.9 (sd 1.0) — concentrated, in the direction mimicry predicts. Two of the top matches are Australian species that co-occur with lyrebirds and are plausible models; the rest are Neotropical and African and cannot be mimicry. The accompanying figure of 83.9 % for "nearest centroid is its own type" is circular by construction, since those syllables are inside those centroids, and is not evidence.

At $n = 31$ this licenses nothing beyond a direction. The action it names is a targeted harvest rather than a method: xeno-canto's lyrebird recordings frequently name the mimicked species in the remarks, so such a harvest returns *labelled* mimic-to-model pairs, and other mimics extend the same set. The reason to pursue it is that it would also supply a fidelity yardstick for the objective problem of Section 6 that owes nothing to our synthesiser and nothing to our cosine.

9.10 What the corpus cannot be asked

Stated as a list, because each item is a boundary rather than a shortcoming to be worked around.

Function, for almost everything. Playback is the only proof, there is none here, and every recording is one bird with no interaction and no receiver. The interface correctly reports "real acoustics, no documented function" for the overwhelming majority of the corpus: at the production build, 13 919 of 14 003 regions are *unlabelled* against 27 *documented*, 10 *experimental*, 16 *inferred* and 31 *invented*. The store holds 23 697 syllables the novelty detector has flagged, of which 841 are reported as *unlabelable* even by an external model, because their xeno-canto audio was deleted by design — a cost of the retention policy, stated rather than absorbed.

Species replication on retained audio. The cache holds at most two recordings per species, so any test needing several recordings of one species requires a re-fetch.

Geographic priors. BirdNET's occurrence meta-model takes latitude, longitude and week, and the store keeps date and country but *no coordinates*. Both archives return them; the harvester never recorded them. That is a metadata-only re-fetch and it is the gate on the only part of the external comparison still unanswered.

Individual identity and dialect. The corpus now has the depth in principle — 10 to 16 individuals per species is literature-proven separable — and the recordist confound of Section 6 makes any such claim uninterpretable until it is addressed in the data.

10 Discussion

Three of this paper's results generalise past the system that produced them.

Recording-condition confounding is a property of the public bioacoustic archives, not of any one model's loss. It has now been attacked from four independent directions — a gradient-reversed adversarial head, a species-factored version of the same, the post-hoc genomics batch-effect toolkit including its strongest member, and two archive-scale external embeddings — and every route either fails to reach it or pays for it one-for-one in species structure. The two external models disagree with each other by more than either differs from ours, which is the decisive form of the argument: if archive-scale training conferred the property, both would have it and they would agree. Since most species in these archives are carried by a handful of recordists, no method that sees only an embedding can separate labels the sample never separated. The prescription is a data action — more recordists, and more genuinely different recording domains, per species — and it now has two independent derivations, one from domain merging and one from batch correction.

An excellent discriminator is not an objective. A critic that separates real from synthetic at AUC 0.990 on entirely held-out species is monotonically *harmful* when its logit is added to the fit objective. Separating two distributions is not the same as having a gradient that points at the nearer one. Retaining a guardrail metric the learned term cannot game is what made this visible rather than a mystery, and that arrangement generalises to any learned-metric-in-the-loop setup.

Small samples flatter the arm you hoped for, in both directions. The project measured this twice, at similar cost. A cross-domain contrastive mechanism worth +13.4 points on 12 species with a 3-species holdout was worth +1.7 on 63 species with 16 held out; in the same re-test the arm it was compared against also shrank, and one of its metrics reversed sign. An objective change winning 7 of 10 types won 25 of 80. In both cases the corrective was a pre-registered re-test at scale rather than an argument about whether the first result was noise.

For anyone building a comparable system, the ordering the evidence supports is: fix the representation before the clustering, fix the corpus before the loss, and do not spend anything on the search's starting point. And write the gate down first, because three of this project's most useful findings are gates that failed in ways that identified the real constraint — the label-assignment arm, whose failure showed that seeded k-means is nearly as good as knowing the answer where the cluster budget does not bind; the inverse encoder, whose failure eliminated the starting point as an explanation; and the second-order syntax arm, whose failure converted a method question into a countable data question.

11 Limitations

Scale of the evidence. A large share of the comparative numbers in this paper are measured on a 100-species debug subset, on a 37-species store, or — for the domain-merge arms — on 12 or 63 species. That is a deliberate operating rule, adopted after a full-corpus build exhausted the machine's memory, and it is marked at every occurrence. The production-scale claims are the corpus counts, the fit medians, the sequence statistics and the built asset set; the arm-versus-arm comparisons are mostly not.

Self-reported metrics. Cluster purity and type coverage are identities rather than measurements under label assignment, and the shipped manifest says so. Type retrieval is the statistic that survives because nothing forces it, and it is the one quoted.

No audible evaluation. There is no listening test, no mean-opinion score and no playback experiment of our own. The fidelity claims rest on mel-cepstral distortion and multi-resolution spectral distance against cached audio, on 200 types, and the objective that produced the fits does not correlate with either. A re-judgment of every cosine-gated adoption on the spectral bench is owed.

Unresolved internal conflicts. T1's verdict rests on a coverage tie-break between two populations that disagree. The fixed-waveguide tract improves everything in isolation, fails across 22 species on one store, and wins under learned typing on another. The three-dimensional projection's distance faithfulness fell from 0.903 to 0.747 when the synthetic anchors were removed, so on-screen distance now means less than it did, even though nothing about the regions or the matching uses those positions.

Production-scale re-measurement, now done, and it costs. The map-agreement decomposition had only been measured on a debug build, and its own note said to re-run it on the production store before any production decision. The production build carries the figures and they are markedly worse than the debug build predicted: region agreement **0.4909** against the debug build's 0.6316, tie-adjusted **0.7056** against 0.8136, and species-level **0.5619** against 0.6831 — between 12 and 14 points lost on every one of the three. The direction is explicable rather than mysterious: agreement is a nearest-centroid lookup, and this build has 14 003 knots where the debug build had 4 298, so a

held-out syllable has three times as many near neighbours available to be confused with the right one. The consequence is the one that matters for the interface's central gesture. On the shipped production build, a live syllable routed through the descriptor map reaches the same region the embedding itself would choose **less than half the time**, and the same species just over half. The tie-adjusted figure says most of the disagreement is a near-miss between adjacent knots rather than an unrelated region, which is the difference between a blurred answer and a wrong one — but it does not make the raw figure larger, and the raw figure is what a reader should hold. Taken with the 0.602 retrieval ceiling of §7, the honest bound on "sing and the right region lights" at production scale is lower than any debug number in this paper.

Unrun. A per-syllable comparison against the external models. A novelty comparison against them. The occurrence prior, which is blocked on coordinates the harvester never recorded. Simulation-based inference and differentiable DSP for the fit. Weak function labels from the archives' own behavioural annotations. Each is specified with its gate in the project's plan document.

12 Conclusion

Lyrebird measures 340 079 syllables from wild recordings without keeping the recordings, fits a published physical syrinx model to 13 859 syllable types over 2 421 species, learns a measured grammar over those types, and displays the result as a map in which live sound lights the acoustically nearest regions — with an evidence label on every claim.

The results that survive scrutiny are narrower than the system. A first-order transition model over the learned tokens buys 0.98 bits per token held out while costing 0.75 on the shuffle, which establishes that the tokens are units. Repeat classes take the corpus from 48 to 98 species with a measured song while the shuffle control does not move. A second order is not supported, and 217 of 2 421 species have enough bouts to ask. Type retrieval tops out at 0.602, so for two syllables in five the nearest region is the wrong one. The objective that produced every fit does not rank audible fidelity. And the embedding organises by recordist before it organises by bird — a property of the archives, established four independent ways, including from outside our own loss entirely.

Thirty-six of fifty-eight pre-registered gates failed. The complete ledger is Appendix A. Those refusals are the part of this work most likely to save someone else time, and they are the reason the positive results above can be stated without hedging.

Reproducibility and data availability

The pipeline, the browser application and the complete research record are at **[TODO: repository URL and archived release DOI]**. The harvest stores are not distributed: they contain no audio, but they were gathered under archive rate limits and are the only copies. Every one-off measurement in Sections 7 and 8 is a committed script whose pre-registered gate is the docstring at the top of the file, on the principle that a result whose gate cannot be re-read is not evidence. Two fixed artefacts the comparative arms depend on — a dumped clustering matrix and a local audio cache — sit outside the repository by design, so absolute numbers will move on regeneration while the arm-against-arm comparisons hold.

Ethics and licensing

All recordings come from xeno-canto and iNaturalist under their per-file licences. Recordings under a NoDerivatives licence are refused at harvest, because every downstream product is a derived measurement, and recordings with no recorded licence are refused, because absence of a licence is not permission. Ninety point three per cent of the production store's 10 184 recordings pass, and all 10 184 carry an attributed recordist. No audio is redistributed and none reaches the browser. External model weights are used offline only, under their own licences, and their outputs are recorded as metadata that no part of the inventory may read. This is a non-commercial research project.

Acknowledgements

[TODO: acknowledgements, funding statement or explicit statement of none]

References

[1] Ana Amador and Gabriel B. Mindlin. Beyond harmonic sounds in a simple model for birdsong production. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 18(4):043123, 2008. doi: 10.1063/1.3041023. URL https://doi.org/10.1063/1.3041023.

[2] Ana Amador and Gabriel B. Mindlin. Synthetic birdsongs as a tool to induce, and listen to, replay activity in sleeping birds. *Frontiers in Neuroscience*, 15:647978, 2021. doi: 10.3389/fnins.2021.647978. URL https://doi.org/10.3389/fnins.2021.647978. Review article; the noise-on-tension result it reproduces originates in amador2013.

[3] Ana Amador, Yonatan Sanz Perl, Gabriel B. Mindlin, and Daniel Margoliash. Elemental gesture dynamics are encoded by song premotor cortical neurons. *Nature*, 495(7439):59–64, 2013. doi: 10.1038/nature11967. URL https://doi.org/10.1038/nature11967.

[4] Ezequiel M. Arneodo, Yonatan Sanz Perl, Franz Goller, and Gabriel B. Mindlin. Prosthetic avian

vocal organ controlled by a freely behaving bird based on a low dimensional model of the biomechanical periphery. *PLOS Computational Biology*, 8(6):e1002546, 2012. doi: 10.1371/journal.pcbi.1002546. URL https://doi.org/10.1371/journal.pcbi.1002546.

[5] Gabriël J. L. Beckers, Roderick A. Suthers, and Carel ten Cate. Pure-tone birdsong by resonance filtering of harmonic overtones. *Proceedings of the National Academy of Sciences*, 100(12):7372–7376, 2003. doi: 10.1073/pnas.1232227100. URL https://doi.org/10.1073/pnas.1232227100. Species are doves (*Streptopelia*), not passerines.

[6] Robert C. Berwick, Kazuo Okanoya, Gabriël J. L. Beckers, and Johan J. Bolhuis. Songs to syntax: the linguistics of birdsong. *Trends in Cognitive Sciences*, 15(3):113–121, 2011. doi: 10.1016/j.tics.2011.01.002. URL https://doi.org/10.1016/j.tics.2011.01.002.

[7] Roberto Bistel, Ana Amador, and Gabriel B. Mindlin. Response of wild songbirds to songs synthesized with a low-dimensional model. *Physical Review E*, 109(5):054410, 2024. doi: 10.1103/PhysRevE.109.054410. URL https://doi.org/10.1103/PhysRevE.109.054410.

[8] Santiago Boari, Yonatan Sanz Perl, Ana Amador, Daniel Margoliash, and Gabriel B. Mindlin. Automatic reconstruction of physiological gestures used in a model of birdsong production. *Journal of Neurophysiology*, 114(5):2912–2922, 2015. doi: 10.1152/jn.00385.2015. URL https://doi.org/10.1152/jn.00385.2015.

[9] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*, volume 7819 of *Lecture Notes in Computer Science*, pages 160–172. Springer, 2013. doi: 10.1007/978-3-642-37456-2_14. URL https://doi.org/10.1007/978-3-642-37456-2_14.

[10] Mustafa Chasmai, Alexander Shepard, Subhransu Maji, and Grant Van Horn. The iNaturalist sounds dataset. In *Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2506.00343. Preprint posted 2025, after the NeurIPS 2024 presentation.

[11] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proc. 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 1597–1607, 2020. URL http://proceedings.mlr.press/v119/chen20j.html.

[12] Rudi Cilibrasi and Paul M. B. Vitányi. Clustering by compression. *IEEE Transactions on Information Theory*, 51(4):1523–1545, 2005. doi: 10.1109/TIT.2005.844059. URL https://doi.org/10.1109/TIT.2005.844059. Preprint arXiv:cs/0312044 (2003). The normalised compression distance.

[13] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020. doi: 10.1073/pnas.1912789117. URL https://doi.org/10.1073/pnas.1912789117.

[14] Anastasia H. Dalziell and Robert D. Magrath. Fooling the experts: accurate vocal mimicry in the song of the superb lyrebird, *menura novaehollandiae*. *Animal Behaviour*, 83(6):1401–1410, 2012. doi: 10.1016/j.anbehav.2012.03.009. URL https://doi.org/10.1016/j.anbehav.2012.03.009. Shrike-thrushes reacted as strongly to mimetic song as to their own when songs were presented *alone*; approach dropped when the mimicry was embedded in lyrebird song sequences.

[15] Coen P. H. Elemans, Andrew F. Mead, Lawrence C. Rome, and Franz Goller. Superfast vocal muscles control song production in songbirds. *PLOS ONE*, 3(7):e2581, 2008. doi: 10.1371/journal.pone.0002581. URL https://doi.org/10.1371/journal.pone.0002581.

[16] Coen P. H. Elemans, Riccardo Zaccarelli, and Hanspeter Herzel. Biomechanics and control of vocalization in a non-songbird. *Journal of the Royal Society Interface*, 5(24):691–703, 2008. doi: 10.1098/rsif.2007.1237. URL https://doi.org/10.1098/rsif.2007.1237. Cited in the repository as 2009, vol. 6, pp. 491–506; all three are wrong.

[17] Jesse Engel, Lamtharn Hantrakul, Chenjie Gu, and Adam Roberts. DDSP: Differentiable digital signal processing. In *Proc. International Conference on Learning Representations (ICLR)*, 2020. URL https://arxiv.org/abs/2001.04643.

[18] Facundo Fainstein, Franz Goller, and Gabriel B. Mindlin. Nonlinear biomechanical resonances in birdsong, 2025. URL https://arxiv.org/abs/2509.11019. Preprint, v2. The v1 title was "Birds breathe and sing at resonances of the biomechanics"; cite the version.

[19] Livio Favaro, Marco Gamba, Eleonora Cresta, Elena Fumagalli, Francesca Bandoli, Cristina Pilenga, Valentina Isaja, Nicolas Mathevon, and David Reby. Do penguins' vocal sequences conform to linguistic laws? *Biology Letters*, 16(2):20190589, 2020. doi: 10.1098/rsbl.2019.0589. URL https://doi.org/10.1098/

rsbl.2019.0589. African penguin. Menzerath–Altmann holds, but by shifting sequence composition toward the shortest syllable type rather than by abbreviating all syllable types as in gustison2016.

[20] Michale S. Fee, Boris Shraiman, Bijan Pesaran, and Partha P. Mitra. The role of nonlinear dynamics of the syrinx in the vocalizations of a songbird. *Nature*, 395(6697):67–71, 1998. doi: 10.1038/25725. URL https://doi.org/10.1038/25725.

[21] Neville H. Fletcher, Tobias Riede, and Roderick A. Suthers. Model for vocalization by a bird with distensible vocal cavity and open beak. *Journal of the Acoustical Society of America*, 119(2):1005–1011, 2006. doi: 10.1121/1.2159434. URL https://doi.org/10.1121/1.2159434.

[22] Todd M. Freeberg, Jeffrey R. Lucas, and Barbara Clucas. Variation in chick-a-dee calls of a carolina chickadee population, *poecile carolinensis*: identity and redundancy within note types. *Journal of the Acoustical Society of America*, 113(4):2127–2136, 2003. doi: 10.1121/1.1559175. URL https://doi.org/10.1121/1.1559175. Carolina chickadee, and about sex/microgeographic identity — not the A-B-C-D syntax and not D-count scaling.

[23] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016. URL https://jmlr.org/papers/v17/15-239.html. The gradient-reversal layer. No DOI (JMLR). Note that JMLR's own exported BibTeX truncates the seventh author to "Mario March"; the correct name is Marchand.

[24] Tim Gardner, Guillermo Cecchi, Marcelo Magnasco, Rodrigo Laje, and Gabriel B. Mindlin. Simple motor gestures for birdsongs. *Physical Review Letters*, 87(20):208101, 2001. doi: 10.1103/PhysRevLett.87.208101. URL https://doi.org/10.1103/PhysRevLett.87.208101.

[25] Jack Goffinet, Samuel Brudner, Richard Mooney, and John Pearson. Low-dimensional learned feature spaces quantify individual and group differences in vocal repertoires. *eLife*, 10:e67855, 2021. doi: 10.7554/eLife.67855. URL https://doi.org/10.7554/eLife.67855.

[26] Franz Goller and Roderick A. Suthers. Role of syringeal muscles in gating airflow and sound production in singing brown thrashers. *Journal of Neurophysiology*, 75(2):867–876, 1996. doi: 10.1152/jn.1996.75.2.867. URL https://doi.org/10.1152/jn.1996.75.2.867.

[27] Michael Griesser. Referential calls signal predator behavior in a group-living bird species. *Current Biology*, 18(1):69–73, 2008. doi: 10.1016/j.cub.2007.11.069. URL https://doi.org/10.1016/j.cub.2007.11.069.

[28] Morgan L. Gustison, Stuart Semple, Ramon Ferrer-i Cancho, and Thore J. Bergman. Gelada vocal sequences follow Menzerath's linguistic law. *Proceedings of the National Academy of Sciences*, 113(19):E2750–E2758, 2016. doi: 10.1073/pnas.1522072113. URL https://doi.org/10.1073/pnas.1522072113.

[29] Masato Hagiwara. AVES: Animal vocalization encoder based on self-supervision. In *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023. URL https://arxiv.org/abs/2210.14493.

[30] Masato Hagiwara, Benjamin Hoffman, Jen-Yu Liu, Maddie Cusimano, Felix Effenberger, and Katie Zacarian. BEANS: The benchmark of animal sounds. In *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2023. doi: 10.1109/ICASSP49357.2023.10096686. URL https://doi.org/10.1109/ICASSP49357.2023.10096686. Preprint arXiv:2210.12300 (2022), which is where the key's year comes from; the published version is ICASSP 2023.

[31] Rebecca Schurr Hartley and Roderick A. Suthers. Airflow and pressure during canary song: direct evidence for mini-breaths. *Journal of Comparative Physiology A*, 165(1):15–26, 1989. doi: 10.1007/BF00613795. URL https://doi.org/10.1007/BF00613795. Mini-breaths at 2–27 notes/s, pulsatile expiration at 30–38 notes/s.

[32] Henrike Hultsch and Dietmar Todt. Repertoire sharing and song-post distance in nightingales (*luscinia megarhynchos* b.). *Behavioral Ecology and Sociobiology*, 8(3):183–188, 1981. doi: 10.1007/BF00299828. URL https://doi.org/10.1007/BF00299828.

[33] Henrike Hultsch and Dietmar Todt. Memorization and reproduction of songs in nightingales (*luscinia megarhynchos*): evidence for package formation. *Journal of Comparative Physiology A*, 165(2):197–203, 1989. doi: 10.1007/BF00619194. URL https://doi.org/10.1007/BF00619194.

[34] Dezhe Z. Jin and Alexay A. Kozhevnikov. A compact statistical model of the song syntax in bengalese finch. *PLOS Computational Biology*, 7(3):e1001108, 2011. doi: 10.1371/journal.pcbi.1001108. URL https://doi.org/10.1371/journal.pcbi.1001108. Preprint arXiv:1011.2998 (2010); the repository cites only the preprint.

[35] Stefan Kahl, Connor M. Wood, Maximilian Eibl, and Holger Klinck. BirdNET: A deep learning

solution for avian diversity monitoring. *Ecological Informatics*, 61:101236, 2021. doi: 10.1016/j.ecoinf.2021.101236. URL https://doi.org/10.1016/j.ecoinf.2021.101236. Model weights CC BY-NC-SA 4.0; the BirdNET-Analyzer code is MIT.

[36] Alireza Kazemi, Mariam Kesba, and Pauline Provini. Realistic three-dimensional avian vocal tract model demonstrates how shape affects sound filtering (*passer domesticus*). *Journal of the Royal Society Interface*, 20(198):20220728, 2023. doi: 10.1098/rsif.2022.0728. URL https://doi.org/10.1098/rsif.2022.0728.

[37] Reinhard Kneser and Hermann Ney. Improved backing-off for m-gram language modeling. In *Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, volume 1, pages 181–184, 1995. doi: 10.1109/ICASSP.1995.479394. URL https://doi.org/10.1109/ICASSP.1995.479394.

[38] Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0. URL https://doi.org/10.1038/s41592-019-0619-0.

[39] Robert Kubichek. Mel-cepstral distance measure for objective speech quality assessment. In *Proc. IEEE Pacific Rim Conference on Communications, Computers and Signal Processing (PACRIM)*, volume 1, pages 125–128, 1993. doi: 10.1109/PACRIM.1993.407206. URL https://doi.org/10.1109/PACRIM.1993.407206.

[40] Rodrigo Laje and Gabriel B. Mindlin. Diversity within a birdsong. *Physical Review Letters*, 89 (28):288102, 2002. doi: 10.1103/PhysRevLett.89.288102. URL https://doi.org/10.1103/PhysRevLett.89.288102. A neural-circuit (RA) model driving the syrinx oscillator; not the source of the dimensionless normal form — see amador2008 and sitt2010.

[41] Rodrigo Laje and Gabriel B. Mindlin. Modeling source-source and source-filter acoustic interaction in birdsong. *Physical Review E*, 72(3):036218, 2005. doi: 10.1103/PhysRevE.72.036218. URL https://doi.org/10.1103/PhysRevE.72.036218.

[42] Rodrigo Laje, Timothy J. Gardner, and Gabriel B. Mindlin. Neuromuscular control of vocalizations in birdsong: A model. *Physical Review E*, 65(5):051921, 2002. doi: 10.1103/PhysRevE.65.051921. URL https://doi.org/10.1103/PhysRevE.65.051921.

[43] Ming Li, Xin Chen, Xin Li, Bin Ma, and Paul M. B. Vitányi. The similarity metric. *IEEE Transactions on Information Theory*, 50(12):3250–3264, 2004. doi: 10.1109/TIT.2004.838101. URL https://doi.org/10.1109/TIT.2004.838101. Preprint arXiv:cs/0111054 (2001). Defines the normalised information distance, which is non-computable because it rests on Kolmogorov complexity; cilibrasi2005's NCD is its computable approximation.

[44] Jiali Lu, Sumithra Surendralal, Kristofer E. Bouchard, and Dezhe Z. Jin. Partially observable markov models inferred using statistical tests reveal context-dependent syllable transitions in bengalese finch songs. *Journal of Neuroscience*, 45(9): e0522242024, 2025. doi: 10.1523/JNEUROSCI.0522-24.2024. URL https://doi.org/10.1523/JNEUROSCI.0522-24.2024. The paper argues parsimony and interpretability, not a categorical win over HMMs.

[45] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction, 2018. URL https://arxiv.org/abs/1802.03426.

[46] Gabriel B. Mindlin and Rodrigo Laje. *The Physics of Birdsong*. Biological and Medical Physics, Biomedical Engineering. Springer, Berlin, Heidelberg, 2005. ISBN 978-3-540-25399-0. doi: 10.1007/3-540-28249-1. URL https://doi.org/10.1007/3-540-28249-1.

[47] Marius Miron, David Robinson, Milad Alizadeh, Ellen Gilsenan-McMahon, Gagan Narula, Emmanuel Chemla, Maddie Cusimano, Felix Effenberger, Masato Hagiwara, Benjamin Hoffman, Sara Keen, Diane Kim, Jane Lawton, Jen-Yu Liu, Aza Raskin, Olivier Pietquin, and Matthieu Geist. AVEX: What matters for animal vocalization encoding. In *Proc. International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2508.11845. Code (MIT) at https://github.com/earthspecies/avex.

[48] Tobias Morocutti, Florian Schmid, Khaled Koutini, and Gerhard Widmer. Device-robust acoustic scene classification via impulse response augmentation. In *Proc. 31st European Signal Processing Conference (EUSIPCO)*, 2023. URL https://arxiv.org/abs/2305.07499. Evaluated on acoustic scene classification across recording devices, not on bioacoustics.

[49] Ilyass Moummad, Romain Serizel, Emmanouil Benetos, and Nicolas Farrugia. Domain-invariant representation learning of bird sounds, 2024. URL https://arxiv.org/abs/2409.08589. Preprint.

[50] Brian S. Nelson, Gabriël J. L. Beckers, and Roderick A. Suthers. Vocal tract filtering and sound radiation in a songbird. *Journal of Experimental Biology*, 208(2):297–308, 2005. doi: 10.1242/jeb.01378. URL https://doi.org/10.1242/jeb.

01378. The gape regression $A = -5 + 32\log_{10}(B)$ is reported at 6.5 kHz only.

[51] Stephen Nowicki. Vocal tract resonances in oscine bird sound production: evidence from birdsongs in a helium atmosphere. *Nature*, 325(6099):53–55, 1987. doi: 10.1038/325053a0. URL https://doi.org/10.1038/325053a0.

[52] Verena R. Ohms, Peter Ch. Snelderwaard, Carel ten Cate, and Gabriël J. L. Beckers. Vocal tract articulation in zebra finches. *PLOS ONE*, 5(7): e11923, 2010. doi: 10.1371/journal.pone.0011923. URL https://doi.org/10.1371/journal.pone.0011923. Reports gape as a passive filter emphasising frequencies above 5 kHz, not a gape–f0 correlation.

[53] Kazuo Okanoya. The bengalese finch: A window on the behavioral neurobiology of birdsong syntax. *Annals of the New York Academy of Sciences*, 1016:724–735, 2004. doi: 10.1196/annals.1298.026. URL https://doi.org/10.1196/annals.1298.026. Review.

[54] Pritty Patel-Grosz, Salvador Mascarenhas, Emmanuel Chemla, and Philippe Schlenker. Super linguistics: an introduction. *Linguistics and Philosophy*, 46(4):627–692, 2023. doi: 10.1007/s10988-022-09377-8. URL https://doi.org/10.1007/s10988-022-09377-8. A methodological programme; it does not itself assert the song-syntax / call-meaning asymmetry.

[55] Yonatan Sanz Perl, Ezequiel M. Arneodo, Ana Amador, Franz Goller, and Gabriel B. Mindlin. Reconstruction of physiological instructions from zebra finch song. *Physical Review E*, 84(5):051909, 2011. doi: 10.1103/PhysRevE.84.051909. URL https://doi.org/10.1103/PhysRevE.84.051909.

[56] Jeffrey Podos, Joel A. Southall, and Marcos R. Rossi-Santos. Vocal mechanics in darwin's finches: correlation of beak gape and song frequency. *Journal of Experimental Biology*, 207(4):607–619, 2004. doi: 10.1242/jeb.00770. URL https://doi.org/10.1242/jeb.00770.

[57] Simon J. D. Prince and James H. Elder. Probabilistic linear discriminant analysis for inferences about identity. In *Proc. IEEE 11th International Conference on Computer Vision (ICCV)*, pages 1–8, 2007. doi: 10.1109/ICCV.2007.4409052. URL https://doi.org/10.1109/ICCV.2007.4409052. A face-recognition paper. PLDA was adopted by the speaker-verification community later; this is the method's origin, not its speaker-ID use.

[58] Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems 20 (NIPS)*, 2007. URL https://papers.nips.cc/paper_files/paper/2007/hash/013a006f03dbc5392effeb8f18fda755-Abstract.html.

[59] Lukas Rauch, René Heinrich, Ilyass Moummad, Alexis Joly, Bernhard Sick, and Christoph Scholz. Can masked autoencoders also listen to birds? *Transactions on Machine Learning Research*, 2025. URL https://arxiv.org/abs/2504.12880.

[60] Lukas Rauch, Raphael Schwinger, Moritz Wirth, René Heinrich, Denis Huseljic, Marek Herde, Jonas Lange, Stefan Kahl, Bernhard Sick, Sven Tomforde, and Christoph Scholz. BirdSet: A large-scale dataset for audio classification in avian bioacoustics. In *Proc. International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2403.10380.

[61] Tobias Riede and Franz Goller. Peripheral mechanisms for vocal production in birds – differences and similarities to human speech and singing. *Brain and Language*, 115(1):69–80, 2010. doi: 10.1016/j.bandl.2009.11.003. URL https://doi.org/10.1016/j.bandl.2009.11.003.

[62] Tobias Riede and Roderick A. Suthers. Vocal tract motor patterns and resonance during constant frequency song: the white-throated sparrow. *Journal of Comparative Physiology A*, 195(2):183–192, 2009. doi: 10.1007/s00359-008-0397-0. URL https://doi.org/10.1007/s00359-008-0397-0.

[63] Tobias Riede, Roderick A. Suthers, Neville H. Fletcher, and William E. Blevins. Songbirds tune their vocal tract to the fundamental frequency of their song. *Proceedings of the National Academy of Sciences*, 103(14):5543–5548, 2006. doi: 10.1073/pnas.0601262103. URL https://doi.org/10.1073/pnas.0601262103.

[64] David Robinson, Marius Miron, Masato Hagiwara, Benno Weck, Sara Keen, Milad Alizadeh, Gagan Narula, Matthieu Geist, and Olivier Pietquin. NatureLM-audio: an audio-language foundation model for bioacoustics. In *Proc. International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2411.07186.

[65] Christian Rutz, Michael Bronstein, Aza Raskin, Sonja C. Vernes, Katherine Zacarian, and Damián E. Blasi. Using machine learning to decode animal communication. *Science*, 381(6654):152–155, 2023. doi: 10.1126/science.adg7314. URL https://doi.org/10.1126/science.adg7314. Perspective.

[66] Tim Sainburg, Marvin Thielk, and Timothy Q. Gentner. Finding, visualizing, and quantifying latent structure across diverse animal vocal repertoires. *PLOS Computational Biology*, 16(10):e1008228, 2020. doi: 10.1371/journal.

pcbi.1008228. URL https://doi.org/10.1371/journal.pcbi.1008228.

[67] Pratyusha Sharma, Shane Gero, Roger Payne, David F. Gruber, Daniela Rus, Antonio Torralba, and Jacob Andreas. Contextual and combinatorial structure in sperm whale vocalisations. *Nature Communications*, 15:3617, 2024. doi: 10.1038/s41467-024-47221-8. URL https://doi.org/10.1038/s41467-024-47221-8. At least 143 frequently realised coda combinations; the figure "156" does not appear in the paper.

[68] Jacobo D. Sitt, Ana Amador, Franz Goller, and Gabriel B. Mindlin. Dynamical origin of spectrally rich vocalizations in birdsong. *Physical Review E*, 78(1):011905, 2008. doi: 10.1103/PhysRevE.78.011905. URL https://doi.org/10.1103/PhysRevE.78.011905.

[69] Jacobo D. Sitt, Ezequiel M. Arneodo, Franz Goller, and Gabriel B. Mindlin. Physiologically driven avian vocal synthesizer. *Physical Review E*, 81(3):031927, 2010. doi: 10.1103/PhysRevE.81.031927. URL https://doi.org/10.1103/PhysRevE.81.031927.

[70] Roderick A. Suthers. Contributions to birdsong from the left and right sides of the intact syrinx. *Nature*, 347(6292):473–477, 1990. doi: 10.1038/347473a0. URL https://doi.org/10.1038/347473a0.

[71] Toshitaka N. Suzuki, David Wheatcroft, and Michael Griesser. Experimental evidence for compositional syntax in bird calls. *Nature Communications*, 7:10986, 2016. doi: 10.1038/ncomms10986. URL https://doi.org/10.1038/ncomms10986. Reversed order (D-ABC) produced a reduced response, not no response.

[72] Evren C. Tumer and Michael S. Brainard. Performance variability enables adaptive plasticity of 'crystallized' adult birdsong. *Nature*, 450(7173):1240–1244, 2007. doi: 10.1038/nature06390. URL https://doi.org/10.1038/nature06390. The "∼1 % f0 CV" figure the repository attributes to this paper could not be located in accessible text.

[73] Bart van Merriënboer, Vincent Dumoulin, Jenny Hamer, Lauren Harrell, Andrea Burns, and Tom Denton. Perch 2.0: The bittern lesson for bioacoustics, 2025. URL https://arxiv.org/abs/2508.04665. 1536-d embedding, EfficientNet-B3 trunk, 14 795 output classes of which 14 597 are species, 5 s at 32 kHz; code Apache 2.0.

[74] William E. Wood, Peter J. Osseward, Thomas K. Roseberry, and David J. Perkel. A daily oscillation in the fundamental frequency and amplitude of harmonic syllables of zebra finch song. *PLOS ONE*, 8(12):e82327, 2013. doi: 10.1371/journal.pone.0082327. URL https://doi.org/10.1371/journal.pone.0082327. True source of the $R^2 \approx 0.03$ figure the repository credits to a non-existent "Chen et al. 2013".

[75] Xeno-canto Foundation. xeno-canto: sharing wildlife sounds from around the world. https://xeno-canto.org. Per-recording Creative Commons licences; accessed 2026-08-01.

[76] Kaiyang Zhou, Yongxin Yang, Yu Qiao, and Tao Xiang. Domain generalization with MixStyle. In *Proc. International Conference on Learning Representations (ICLR)*, 2021. URL https://arxiv.org/abs/2104.02008. Extended journal version: Zhou et al., *MixStyle Neural Networks for Domain Generalization and Adaptation*, IJCV 132(3):822–836, 2024, https://doi.org/10.1007/s11263-023-01913-8.

[77] Sue Anne Zollinger, Tobias Riede, and Roderick A. Suthers. Two-voice complexity from a single side of the syrinx in northern mockingbird *mimus polyglottos* vocalizations. *Journal of Experimental Biology*, 211(12):1978–1991, 2008. doi: 10.1242/jeb.014092. URL https://doi.org/10.1242/jeb.014092.

A The complete pre-registered gate ledger

Table 10 lists every gate the project registered before its run, the outcome, and the verdict. It is complete: refuted arms are not omitted, and arms whose machinery was subsequently deleted keep their row, because the measurement is what survives a deleted flag. Of 58 gates, 20 passed, 36 ended in a written refusal, and 2 are unrun or pending. Four refusals still produced an asset: the map-flip lever became an honesty gate, the mel target became the verdict instrument for future fit gates, label assignment became a build flag, and the source-axis deflation produced the finding that the recorded/synthesised difference is per-type geometry rather than one direction.

Where a row is marked *debug* the measurement is on the 100-species debug subset, the 37-species store, or a similarly sized sample; those numbers license arm-against-arm comparison and not absolute claims.

Table 10: The complete ledger of pre-registered gates.

what was tried	the gate, set in advance	what it measured	verdict
Typing in the critic's 32-d embedding	placement up, cosine guardrail held	38.4 % → 51.7 %, cosine 0.807 → 0.796 *(debug)*	**ships**
Parallel and streaming learn	bit-identical to serial	bit-identical, ∼3 h → ∼15 min	**ships**
Fricative route: noise through fitted formants	resynthesis beats the tonal fit, per type	bench 0.584 → 0.656, 13 wins / 0 losses	**ships per type**
Tracheal comb per species	comb improves the resynthesis	won 10 of 22 species	**ships where it wins**
Measured texture on authored one-tone calls	gain against the same bird's learned types	14 of 15, median +0.148	**ships**
Respiratory oscillator, hand-set default	beat the envelope on median cosine	0.343 → 0.336	fails, parked
Respiratory oscillator, per-species fit	win the type, gate held everywhere	won 445 of 12 680 scored types	**ships per species**
Two-voice coupling, hand-set default	beat the single voice	0.343 → 0.294	fails, parked
Two-voice coupling, per-species fit	win the type	won 2 027 of 5 922 screened	**ships per species**
Fixed-waveguide tract	placement and cosine both up	+6.2 pts on 5 species; regression over 22 (23.1 → 22.2 %)	refuted, gated off
Measured partial series as fit evidence	cosine and placement up	cosine 0.776 → 0.768, placement 61.7 → 55.6 %	refuted
Global spectral-tilt correction	placement up	tilt corrected exactly, placement 23.1 → 20.2 %	refuted
Realism logit as the fit objective	placement and cosine hold	cosine 0.762 → 0.335, placement 17.3 %; replicated at 40 k	**refuted twice**
Mel-path objective at $w = 0.5$	≥ 60 % MCD wins, ΔMCD > +1 dB, cosine drop ≤ 0.02 / max 0.05	31 % wins ($n = 80$), 0.0 dB, max drop 0.120	fails, not adopted
Mel-path as a *measurement*	$\lvert r\rvert \geq 0.4$ against MCD	Spearman −0.420, Pearson −0.305 ($n = 200$)	**adopted as the bench**
Path-cosine mel form	see the damaged-half fixture	drop 0.017 against exp(−LSD)'s 0.540	refuted
RFF ridge browser bridge	median held-out cosine ≥ 0.8	0.962 on the debug build	**ships**
Linear ridge bridge, the incumbent	the same 0.8 gate	0.7997 after the embedding grew finer	refuted, replaced
Source-split cluster seeds	type coverage up, purity no worse	coverage 71.0 → 93.1 % on the dumped matrix, 94.6 % in the build; purity 48.1 → 56.6 %	**ships**
Free extra cluster seeds	the same	coverage 71.0 → 78.8 % (+144 seeds) and 86.9 % (2× types)	superseded by the above
Chunk mining evidence gate	PMI $\cdot\sqrt{\text{count}} \geq 6$, shuffled control finds nothing	control finds 0 chunks; aliasing faked PMI to 1.2 without the gate	**ships**
Corpus-breadth ingest vs the four placement gates	species > 35 %, type ≥ 56 %, cosine ≥ 0.74, $\lvert f_0\rvert \leq 5$ %	species 24.6 → 11.5 % (chance 2.8 → 0.9 %); other three held	**fails** on species
Channel-EQ + Freq-MixStyle augmentation	−30 % recordist lift, species held	−0.7 % recordist *(debug)*	fails
Recordist-adversarial head	the same	−33 % recordist for −14.7 pts species accuracy *(debug)*	fails
Within-species adversarial head	the same, species-safe by construction	species flat to +10 %, recordist −6 % *(debug)*	fails
Naive two-domain merge	cross-domain retrieval ≥ +10 pts	+14.4 pts at 12 species, +3.6 at 63	no adoption; direction holds
Cross-domain contrastive positives	≥ +10 pts over the merge	+13.4 at 12 species, +1.7 at 63	fails on re-test
Centre / whiten by recordist	recordist lift halved, species not falling	recordist 0.1665 → 0.1687 and 0.1540; species −32 to −42 %	refuted

continued overleaf

Table 10, continued

what was tried	the gate, set in advance	what it measured	verdict
Harmony by recordist	the same	recordist −71 %, species −70 %	fails, 1:1 trade
Correction by recording	the same	both removed (−74 / −75 %); labels nearly nested	fails, uninformative
T1: external space vs the recordist confound (Perch)	recordist lift −30 % relative	0.0891 → 0.0744, i.e. −16.5 %	**fails**
T1 with BirdNET	the same	0.0891 → 0.1107, i.e. +24 %	**fails harder**
T3: external species retrieval (Perch)	beat the critic's species lift	0.0074 → 0.0588 (8×)	**passes**
T3 with BirdNET	the same	0.0074 → 0.0393 (5.3×)	passes
T2: external novelty AUC	beat 0.846 / TPR 32.2 %	not run	unrun
Jin–Kozhevnikov repeat-count grammar	beat first-order on held-out bouts	−0.348 nats/token, 0 of 30 species *(debug)*	refuted
Hidden-state HMM grammar	the same	+0.422 nats overall, all of it in the 10 % unseen tail *(debug)*	refuted on decomposition
Kneser–Ney unseen-transition backoff	gain on unseen contexts, no loss on seen	+1.90 / +2.32 nats, seen side exactly 0.0000	**ships**
HMM re-run after the depth harvest	$\geq +0.1$ nats over the repaired first-order	$\approx +0.37$ nats, 78 of 104 species	wins the rule; **not adopted**
Song-level edit-distance kernel	beat bag-of-types on species lift, recordist lift < 0.05	+0.877 vs +0.924; recordist +0.264	fails both gates
Menzerath–Altmann law on the inventory	separate the corpus from its own shuffle	median exponent +0.019 where the law needs negative	the law does not hold
Normalised compression distance	the same	control separates species better (0.038 vs 0.018)	refuted
Second-order structure, plug-in estimator	k=1 → 2 drop above control's	invalid comparison; curves start 2.4 bits apart	retracted by the project
Second-order structure, held-out CE	median $\geq +0.05$ bits/token, $\geq +0.05$ over shuffle, share $\geq$ control +20 pts	−0.091 bits/token; share 25.3 % vs 4.1 %	**fails** (magnitude); share passes
Song-type detection vs a permutation null	recurrence above the null, control ≤ 0.2 %	control 0.170 % at the shipped constants	**ships**
Repeat classes	≥ 90 species with a song, control ≤ 0.2 %	98 species, 356 types, control 0.170 %	**adopted**
Per-species recurrence bar	control returns to 0.2 %	no candidate threshold reaches it (0.266–0.442 %)	refuted, removed
Per-species bout rule	the same	105 species at 0.628 % control yield	refuted, removed
Session gate: off-manifold 5-NN detection	AUC ≥ 0.90	0.953, and ≥ 0.92 from 256 shipped anchors	**ships**
Session gate: map-flip lever	+5 points of agreement	+3.5 points raw; tie-adjusted +8.7 to +13.7	fails; **ships as an honesty gate**
Session gate: prune bad sessions from the fit	evaluation cosine cost ≤ 0.005	−0.0147 and −0.0062 on two stores	refuted
Fewer rendered member variants	seeds per type toward 1.0, purity and coverage no worse	2.000 → 1.993; purity 0.423 → 0.568	fails on the stated gate
Source-axis deflation	the same	42 % of the space's energy removed, source gap 7.233 → 0.0, seeds/type 1.999	refuted
No synthetic point for a type with members	the same	seeds/type exactly 1.000, k 4 298 → 2 155; retrieval +0.02	**adopted, on size**
Label assignment vs uncapped k-means	type retrieval $\geq k$-means +10 pts	+2.5 pts uncapped; +14.2 at production's shed ratio	**fails as written**; ships as a flag
Learned inverse, first bake-off	2× faster at equal cosine	0.737 vs 0.734, and 0.2 s either way	fails its ship rule
Learned inverse as an initialiser	median +0.02, ≥ 60 % of types improved, no species worse than −0.05	+0.0135, 52.8 %, worst species −0.0816	**fails on all three**
Two-start descent	median +0.02 with *no* species worse at all	+0.0330; 0 of 1 661 types and 0 of 100 species worse; cost 61 min vs 11 *(debug)*	**passes**
first attempt, selection before the fricative refinement	the same	+0.0320 but 48 types worse (to −0.049), 100 % of them fricative-flagged	failed, bug found

B Corpus, stores and the built asset set

The current production build holds 348 461 points in 14 003 regions — 13 903 type clusters with zero source splits, plus 100 k-means clusters for the 20 241 points that carry no label — at 100 % type coverage, built in 1 136.8 s with every verification check passing. Its three-dimensional projection reproduces high-dimensional distances at Pearson 0.691 and Spearman 0.689 over 200 000 sampled pairs. Centroid assets are 7.69 MB of JSON plus a 3.81 MB binary sidecar, against the 44 MB whole-file parse that blocked an earlier build. Cluster purity reads 94.5 % and is *not* a measurement on this build — it is 1.0 by construction on the labelled clusters, and the manifest says so.

A separate scaling result belongs here because it is a reproducible engineering fact rather than a finding about birds. The cross-species relations document is quadratic in species count: at 2 421 species it is 2 929 410 unordered pairs and 760 MB, of which only 12 762 pairs (0.44 %) carry a shared type at all. The shipped form writes each species' eight nearest partners, deduplicated — 14 425 pairs, 5.6 MB, a 135-fold reduction — and was checked against the complete document across all 2 421 species at the application's query limit with zero mismatches. The file declares the retention rule and both totals, and states in as many words that the kept shared-type count is not the corpus's shared-type count.

C The oscillator, and the parameters that are fitted

The source is the normal-form flapping oscillator, integrated by forward Euler at 16 to 64 times the audio rate:

$$\dot{x} = y, \tag{1}$$

$$\dot{y} = -\alpha\gamma^2 - \beta\gamma^2 x - \gamma^2 x^3 - \gamma x^2 y + \gamma^2 x^2 - \gamma x y, \tag{2}$$

with x the labial position, y the labial velocity, α the air-sac pressure, β the labial tension and γ a pure time-scale. This is the published normal form term for term, with a sign convention on x that places the limit cycle at $\alpha, \beta > 0$ where the literature states the onset threshold as $\alpha < 0$; the equivalence was verified numerically. Both implementations start from $(x, y) = (10^{-4}, 0)$ and bounds-check the state once per output sample rather than clamping every integration step, because a tight per-step clamp flattens the peaks of the large-amplitude regimes the model legitimately reaches at high tension.

Because rescaling time by γ leaves the equations unchanged, γ is a pure time-scale and

$$f_0 = \gamma\,\Phi(\alpha, \beta), \qquad \Phi(0.1, 1) = 0.17061 \text{ (measured)}, \tag{3}$$

so the calibration table is measured once at $\gamma = 24\,000$ and read at any other γ by scaling the target frequency. Per syllable, $\gamma = \mathrm{clamp}(f_{0,\mathrm{ref}}/0.17061,\ 1200,\ 150\,000)$ with $f_{0,\mathrm{ref}}$ the geometric mean of the syllable's contour frequencies.

The tract is a two-pole resonance centred on the instantaneous f_0, plus up to four fixed resonances at absolute frequencies from the tracheal comb

$$\begin{aligned} f_n &= (2n-1)f_1, \quad n = 1 \ldots 4, \\ \text{gain} &\propto 1/(2n-1), \quad Q = 6, \end{aligned} \tag{4}$$

a stopped quarter-wave series with f_1 measured once per species. Turbulence is added to the source before the tract, scaled by airflow. Roughness multiplies *tension*,

$$\beta \leftarrow \beta \cdot \mathrm{clamp}(1 + 0.12\,\rho\, w\, G,\ 0.5,\ 1.5)\,, \tag{5}$$

with w a xorshift32 noise sequence one-pole lowpassed to 140 Hz and G normalising it to unit standard deviation. Acting on tension rather than on pressure or on the output is the one point the literature is specific about [2], and it makes the frequency and amplitude wobble arrive correlated through the oscillator's own dynamics, which is what separates a rough voice from a clean voice with hiss added.

The optional respiratory path replaces the designed envelope with the published two-state air-sac dynamics [18],

$$\tau_x \dot{x}_s = -(1 + x_s^2)x_s - p + F_0 f(t), \tag{6}$$

$$\tau_p \dot{p} = -(1 + x_s^2)x_s - \big(1 + \alpha_r(p)\big)p + F_0 f(t), \tag{7}$$

with x_s the air-sac wall displacement, p the gauge pressure, α_r a Heaviside conductance giving inflow and outflow different damping, and the tract driven by $\sqrt{p_s}\,y$ — the inertive low-frequency limit of Laje and Mindlin [41], not the acceleration injection some reference implementations use. It is off by default.

Thirteen gesture axes are searched per type by coordinate descent; the measured pitch contour and duration are excluded from the search, because the pitch tracker reaches them to about 1 % and a search must not overwrite a measurement.

Table 11: The measurement stores. Each row is a fixed corpus, and arms are comparable only within a row. The last two rows are the current production state.

store or phase	syllables	species	what it added
original harvest	39 976	22	the reference corpus for the naturalness work
first patch-carrying store	9 041	25	the learned-typing bake-off store
the same at 40 k	39 993	25, 13 after a fixed trim bug	the +13.3-point replication
+ multi-source ingest	39 874	121	the first non-xeno-canto fetch
+ iNatSounds	226 264	3 811	6 453 recordings in one pass, zero failures
+ **depth harvest**	**340 079**	**3 811**	+139 440 syllables, 7 992 → 10 184 recordings
current fit	**327 164 typed**	**2 421 fitted, 13 859 types**	median resynthesis cosine 0.648, 23 minutes